Analog Electronics

Previous Year Questions

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Analog Electronics (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

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- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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UNIT

I

Diode Circuits and Applications

Topics

1. Diode Models with open and short circuit tests
2. Operating Point and Small Signal Analysis of a Diode
3. Zener Diode as a Voltage Regulator
4. Series and Shunt Clipper Circuits
5. Two-Level Clipper Circuits
6. Clamper Circuits and their examples
7. Peak Detector and Average Diode Current
8. Voltage Multiplier Circuits
9. Half Wave Rectifier and Centre Tapped Full Wave Rectifier
10. Full Wave Bridge Rectifier and Capacitive Filters

Unit Snapshot*

Stream: EE	**Questions:** 39	**MCQ:** 29	**MSQ:** 1	**NAT:** 9
1M: 25	**2M:** 14	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2022)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 1M

Consider the circuit shown. Assume that the diode D is ideal. The supply voltage $v_s = 325\sin(2\pi 50t)$ V, $L = 500$ μH, and $R = 10$ Ω. The peak diode current (in amperes) is _____ (Round off to one decimal place)

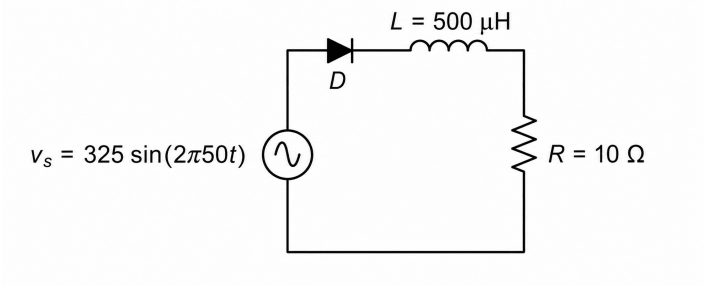


Fig. Q1.26.1

Q1.26.2 NAT 2026 2M

Consider the circuit shown. Assume that the diode (D) is ideal.

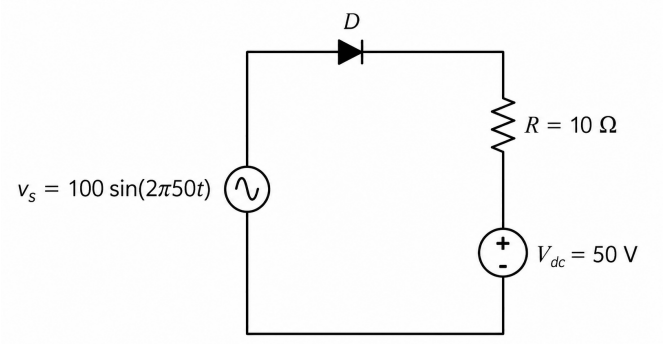


Fig. Q1.26.2

Given $v_s = 100\sin(2\pi 50t)$ V, $V_{dc} = 50$ V and $R = 10$ Ω, the average value of the current through the diode is _____ A (Round off to two decimal places)

Q1.24.1 NAT 2024 2M

In the given circuit, the diodes are ideal. The current through the diode $D1$ in milliamperes is _____ (rounded off to two decimal places).

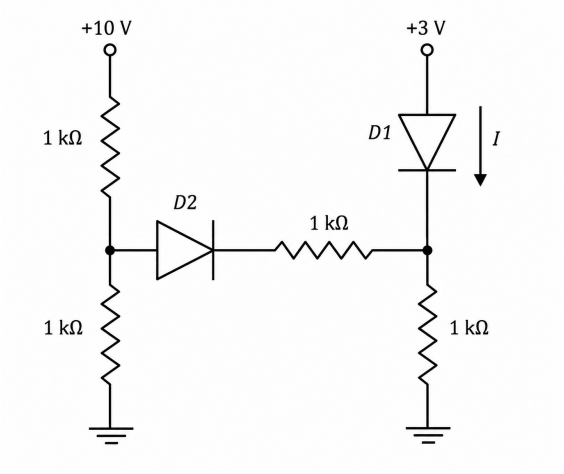


Fig. Q1.24.1

Q1.23.1 MSQ 2023 2M

All the elements in the circuit shown in the following figure are ideal. Which of the following statements is/are true?

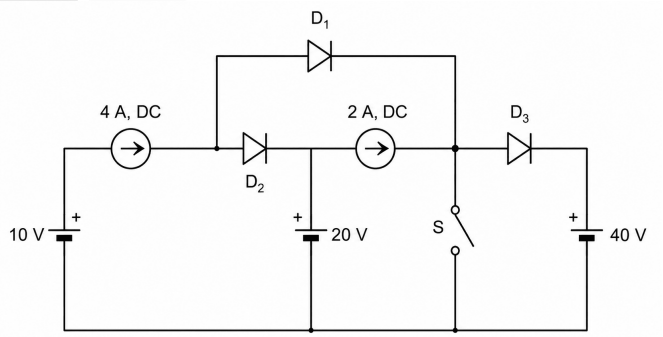


Fig. Q1.23.1

- **A.** When switch S is ON, both D_1 and D_2 conducts and D_3 is reverse biased
- **B.** When switch S is ON, D_1 conducts and both D_2 and D_3 are reverse biased
- **C.** When switch S is OFF, D_1 is reverse biased and both D_2 and D_3 conduct
- **D.** When switch S is OFF, D_1 conducts, D_2 is reverse biased and D_3 conducts

Q1.22.1 MCQ 2022 1M

For the circuit shown below with ideal diodes, the output will be

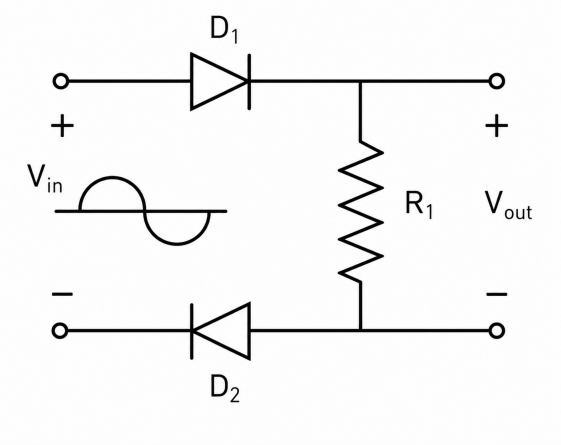


Fig. Q1.22.1

A. $V_{out} = V_{in}$ for $V_{in} > 0$

B. $V_{out} = V_{in}$ for $V_{in} < 0$

C. $V_{out} = -V_{in}$ for $V_{in} > 0$

D. $V_{out} = -V_{in}$ for $V_{in} < 0$

Q1.21.1 NAT 2021 2M

The waveform shown in solid line is obtained by clipping a full-wave rectified sinusoid (shown dashed). The ratio of the RMS value of the full-wave rectified waveform to the RMS value of the clipped waveform is _____. (Round off to 2 decimal places.)

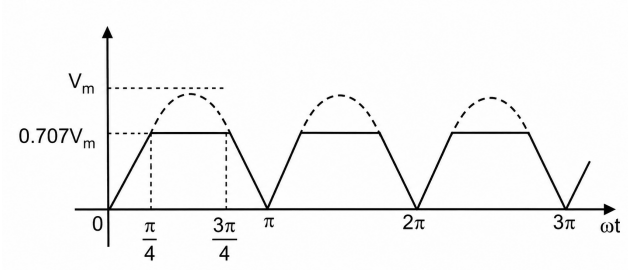


Fig. Q1.21.1

Q1.21.2 NAT 2021 1M

In the circuit shown, a 5 V Zener diode is used to regulate the voltage across load R_0. The input is an unregulated DC voltage with a minimum value of 6 V and a maximum value of 8 V. The value of R is 6 Ω. The Zener diode has a maximum rated power dissipation of 2.5 W. Assuming the Zener diode to be ideal, the minimum value of R_0 is _____ Ω.

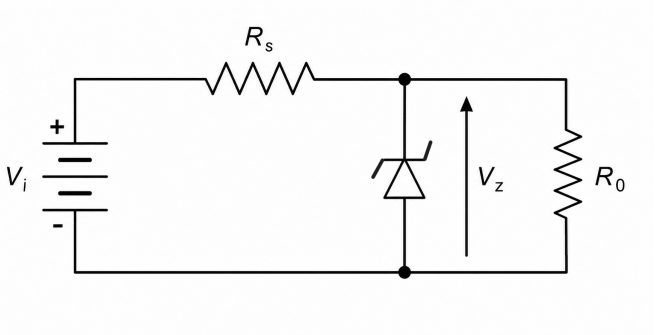


Fig. Q1.21.2

Q1.21.3 NAT 2021 1M

In the circuit shown, the input V_i is a sinusoidal AC voltage having an RMS value of 230V ± 20%. The worst-case peak-inverse voltage seen across any diode is _____ V. (Round off to 2 decimal places.)

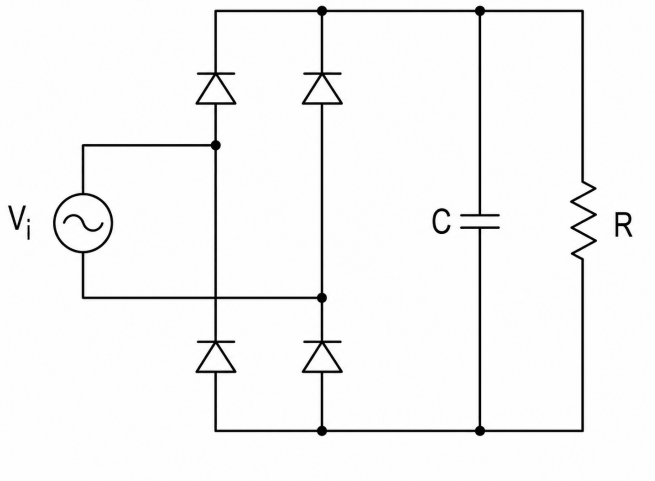


Fig. Q1.21.3

Q1.20.1 MCQ 2020 2M

A non-ideal diode is biased with a voltage of −0.03 V, and a diode current of I_1 is measured. The thermal voltage is 26 mV and the ideality factor for the diode is 15/13. The voltage, in V, at which the measured current increases to $1.5I_1$ is closest to:

A. −0.02 **B.** −0.09 **C.** −1.50 **D.** −4.50

Q1.20.2 MCQ 2020 2M

Consider the diode circuit shown below. The diode, D, obeys the current-voltage characteristic

$$I_D = I_S\left(\exp\left(\frac{V_D}{nV_T}\right) - 1\right),$$

where $n > 1$, $V_T > 0$, V_D is the voltage across the diode and I_D is the current through it. The circuit is biased so that voltage, $V > 0$ and current, $I < 0$. If you had to design this circuit to transfer maximum power from the current source (I_1) to a resistive load (not shown) at the output, what values of R_1 and R_2 would you choose?

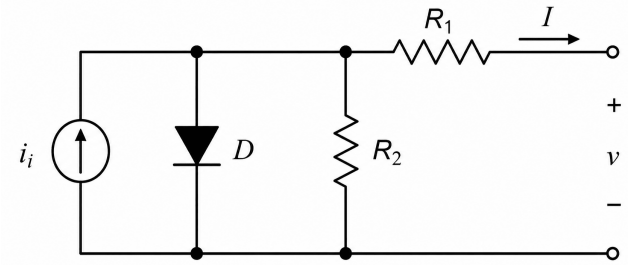


Fig. Q1.20.2

A. Large R_1 and large R_2.

B. Small R_1 and small R_2.

C. Large R_1 and small R_2.

D. Small R_1 and large R_2.

Q1.17.1 MCQ 2017 1M

For the circuit shown in the figure below, assume that diodes D_1, D_2 and D_3 are ideal.

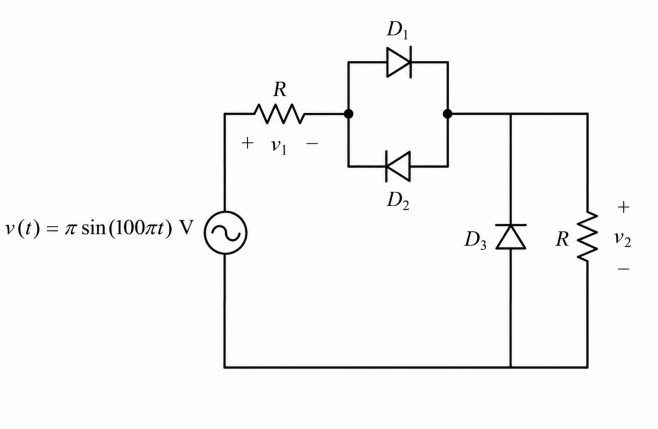


Fig. Q1.17.1

The DC components of voltages v_1 and v_2, respectively are

A. 0 V and 1 V **B.** −0.5 V and 0.5 V
C. 1 V and 0.5 V **D.** 1 V and 1 V

Q1.17.2 MCQ 2017 1M

In the circuit shown, the diodes are ideal, the inductance is small, and $I_0 \neq 0$. Which one of the following statements is true?

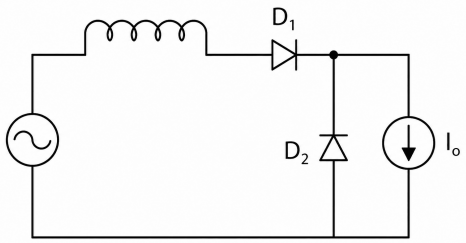


Fig. Q1.17.2

A. D_1 conducts for greater than 180° and D_2 conducts for greater than 180°.

B. D_2 conducts for more than 180° and D_1 conducts for 180°.

C. D_1 conducts for 180° and D_2 conducts for 180°.

D. D_1 conducts for more than 180° and D_2 conducts for 180°.

Q1.15.1 MCQ 2015 1M

In the following circuit, the input voltage V_{in} is $100\sin(100\pi t)$. For $100\pi RC = 50$, the average voltage across R (in volts) under steady-state is nearest to

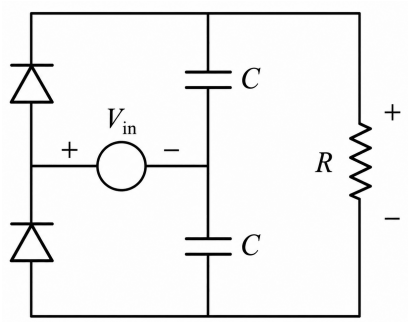


Fig. Q1.15.1

A. 100 **B.** 31.8 **C.** 200 **D.** 63.6

Q1.14.1 MCQ 2014 2M

Assuming the diodes to be ideal in the figure, for the output to be clipped, the input voltage V_i must be outside the range

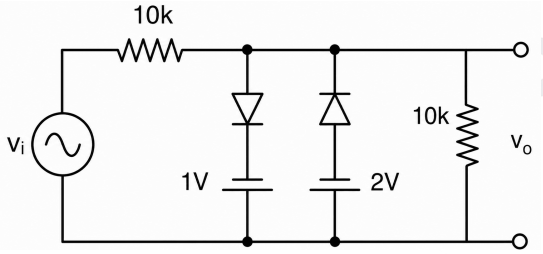


Fig. Q1.14.1

A. −1V to −2V **B.** −2V to −4V
C. +1V to −2V **D.** +2V to −4V

Q1.14.2 NAT 2014 1M

The sinusoidal ac source in the figure has an RMS value of $\frac{20}{\sqrt{2}}$V. Considering all possible values of R_L, the minimum value of R_s in Ω to avoid burnout of the Zener diode is. _____.

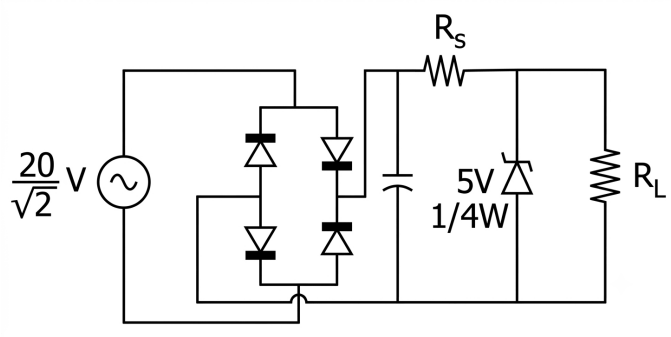


Fig. Q1.14.2

Q1.14.3 NAT 2014 1M

The figure shows the circuit of a rectifier fed from a 230-V (rms), 50-Hz sinusoidal voltage source. If we want to replace the current source with a resistor so that the rms value of the current supplied by the voltage source remains unchanged, the value of the resistance (in ohms) is _____ (Assume diodes to be ideal.)

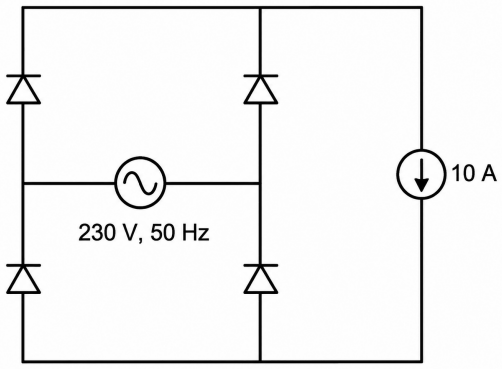


Fig. Q1.14.3

Q1.13.1 MCQ 2013 2M

In the circuit shown below, the knee current of the ideal Zener diode is 10mA. To maintain 5 V across R_L, the minimum value of R_L in Ω and the minimum power rating of the Zener diode in mW respectively are

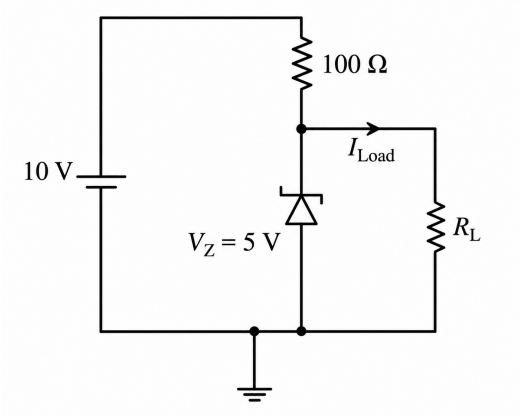


Fig. Q1.13.1

A. 125 and 125 **B.** 125 and 250
C. 250 and 125 **D.** 250 and 250

Q1.13.2 MCQ 2013 1M

A voltage $1000 \sin \omega t$ Volts is applied across YZ. Assuming ideal diodes, the voltage measured across WX in Volts, is

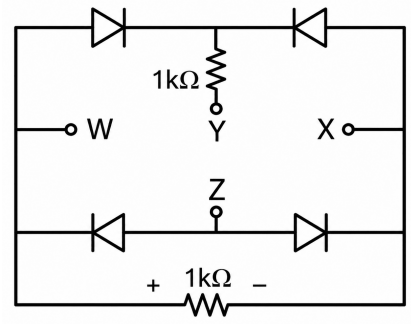


Fig. Q1.13.2

A. $\sin \omega t$ **B.** $(\sin \omega t + |\sin \omega t|)/2$
C. $(\sin \omega t - |\sin \omega t|)/2$ **D.** 0 for all t

Q1.12.1 MCQ 2012 1M

The i-v characteristics of the diode in the circuit given below are

$$i = \begin{cases} \dfrac{v - 0.7}{500}\ \text{A}, & v \geq 0.7\text{V} \\ 0\ \text{A}, & v < 0.7\text{V} \end{cases}$$

The current in the circuit is

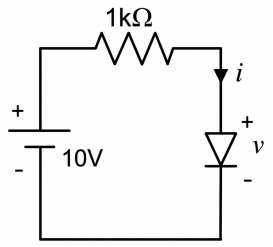


Fig. Q1.12.1

A. 10mA **B.** 9.3mA **C.** 6.67mA **D.** 6.2mA

Q1.11.1 MCQ 2011 2M

A clipper circuit shown below.

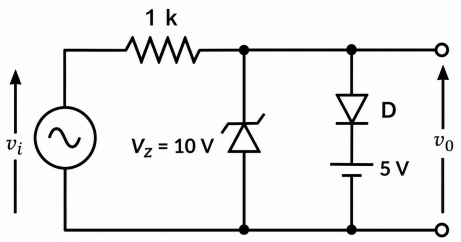
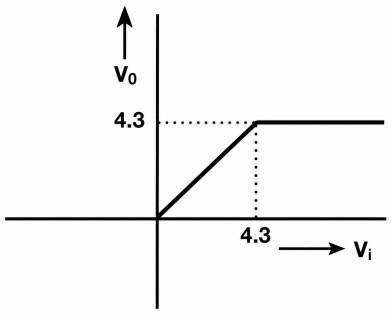


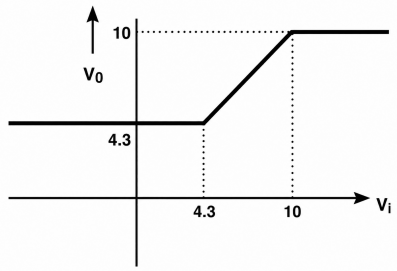
Fig. Q1.11.1

Assuming forward voltage drops of the diodes to be 0.7V, the input-output transfer characteristics of the circuit is

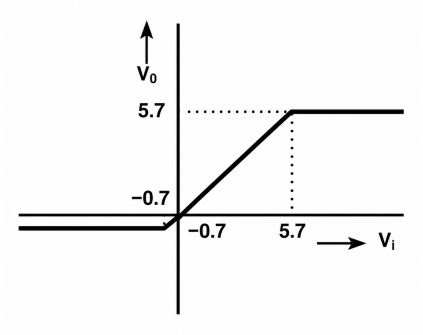
A.



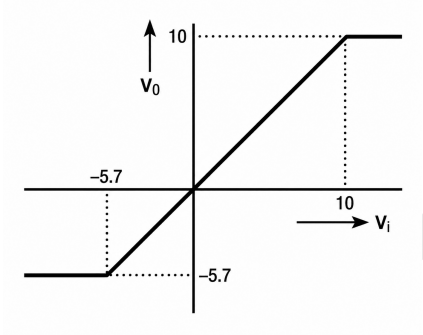
B.



C.



D.



Q1.10.1 MCQ 2010 1M

Assuming that the diodes in the given circuit are ideal, the voltage V_o is

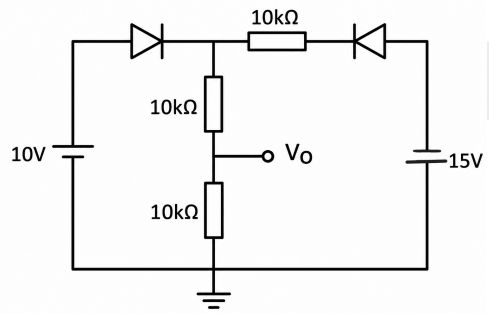


Fig. Q1.10.1

A. 4V **B.** 5V **C.** 7.5V **D.** 12.12V

Q1.09.1 MCQ 2009 1M

The following circuit has a source voltage V_S as shown in the graph. The current through the circuit is also shown.

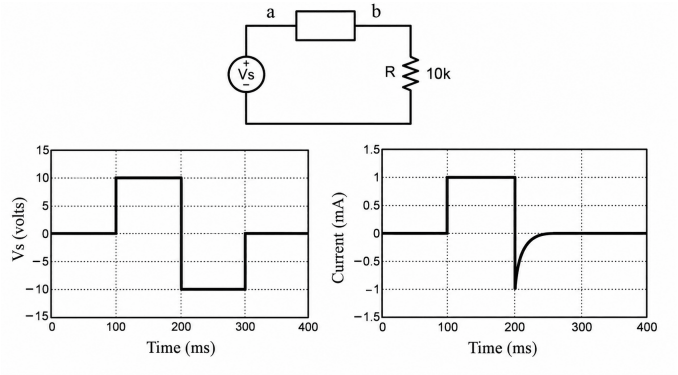
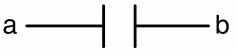
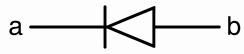


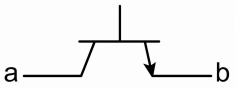
Fig. Q1.09.1

The element connected between **a** and **b** could be

A.

B. 

C. 

D. 

Q1.08.1 MCQ 2008 1M

In the voltage doubler circuit shown in the figure, the switch 'S' is closed at $t = 0$. Assuming diodes D_1 and D_2 to be ideal, load resistance to be infinite and initial capacitor voltages to be zero. The steady state voltage across capacitor C_1 and C_2 will be

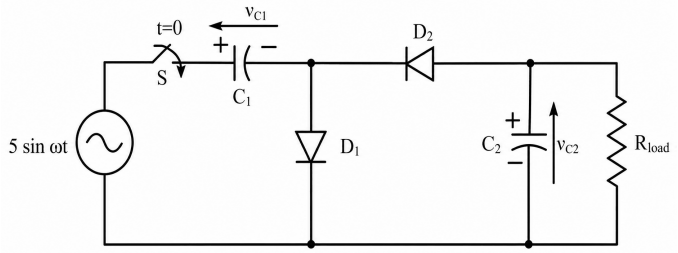


Fig. Q1.08.1

A. $V_{c1} = 10V,\ V_{c2} = 5V$ **B.** $V_{c1} = 10V,\ V_{c2} = -5V$

C. $V_{c1} = 5V,\ V_{c2} = 10V$ **D.** $V_{c1} = 5V,\ V_{c2} = -10V$

Q1.08.2 MCQ 2008 1M

The equivalent circuits of a diode, during forward biased and reverse biased conditions, are shown in the figure.

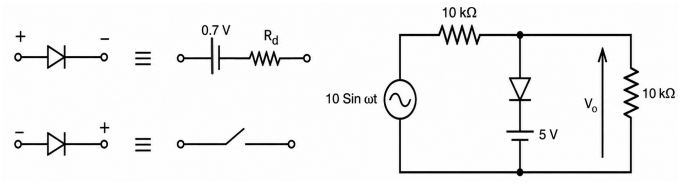
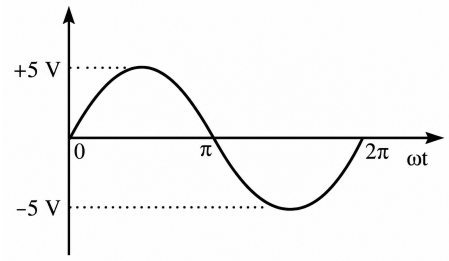


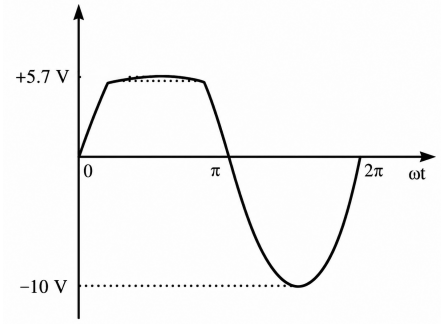
Fig. Q1.08.2

If such a diode is used in clipper circuit of figure given above, the output voltage v_0 of the circuit will be

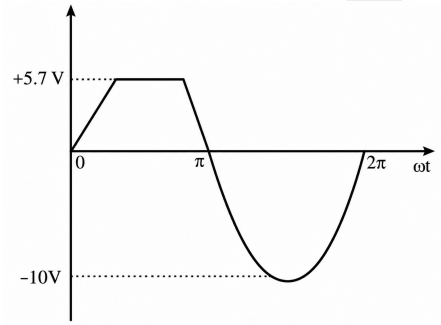
A.



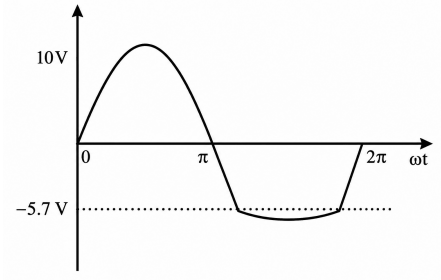
B.



C.



D.



Q1.06.1 MCQ 2006 2M

Assuming the diodes D_1 and D_2 of the circuit shown in figure to be ideal ones, the transfer characteristics of the circuit will be

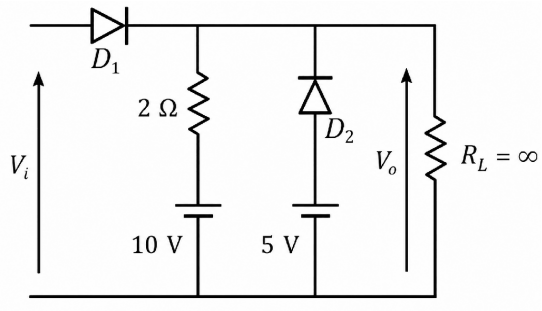
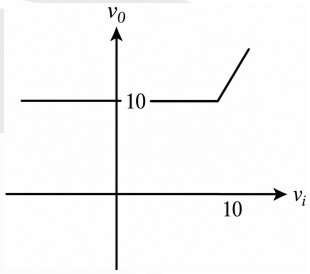
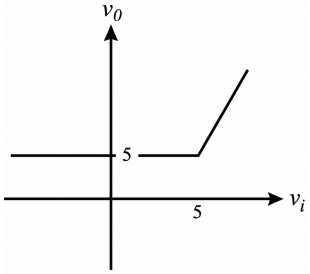


Fig. Q1.06.1

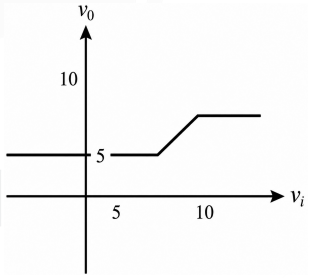
A.



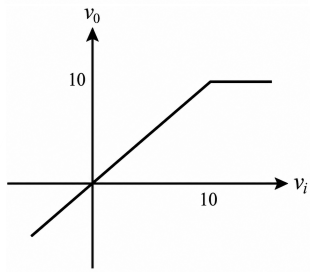
B.



C.



D.



Q1.06.2 MCQ 2006 1M

What are the states of the three ideal diodes of the circuit shown in figure?

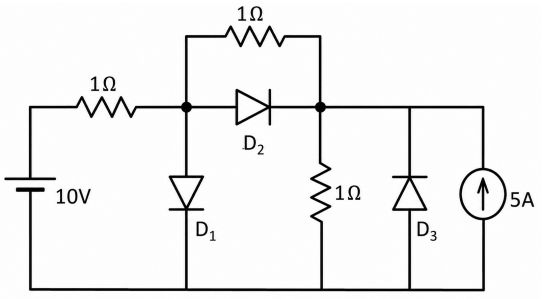


Fig. Q1.06.2

A. D_1 ON, D_2 OFF, D_3 OFF

B. D_1 OFF, D_2 ON, D_3 OFF

C. D_1 ON, D_2 OFF, D_3 ON

D. D_1 OFF, D_2 ON, D_3 ON

Q1.05.1 MCQ 2005 1M

Assume that D_1 and D_2 in Figure are ideal diodes. The value of current I is:

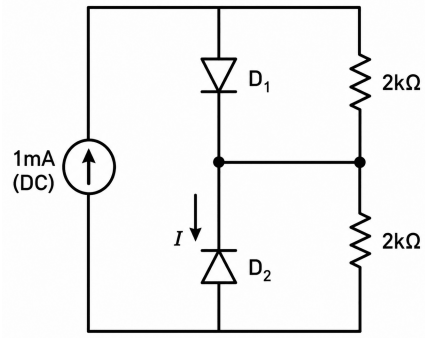


Fig. Q1.05.1

A. 0 mA **B.** 0.5 mA **C.** 1 mA **D.** 2 mA

Q1.04.1 MCQ 2004 2M

Assuming that the diodes are ideal in figure, the current in D_1 is

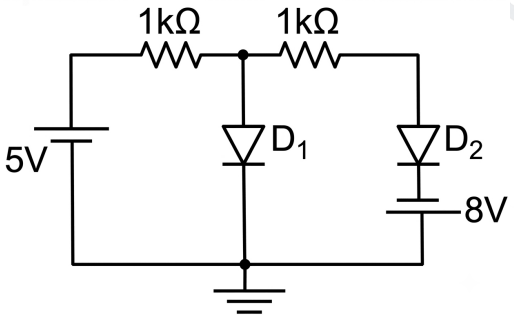


Fig. Q1.04.1

A. 8 mA **B.** 5 mA **C.** 0 mA **D.** −3 mA

Q1.04.2 MCQ 2004 1M

The circuit in figure shows a full-wave rectifier. The input voltage is 230V (RMS) single-phase ac. The peak reverse voltage across the diodes D_1 and D_2 is

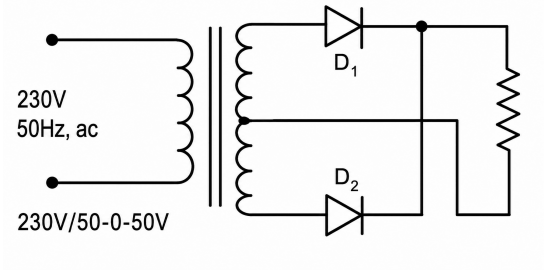


Fig. Q1.04.2

A. $100\sqrt{2}$ V **B.** 100 V **C.** $50\sqrt{2}$ V **D.** 50 V

Q1.04.3 MCQ 2004 1M

The current through the Zener diode in figure is

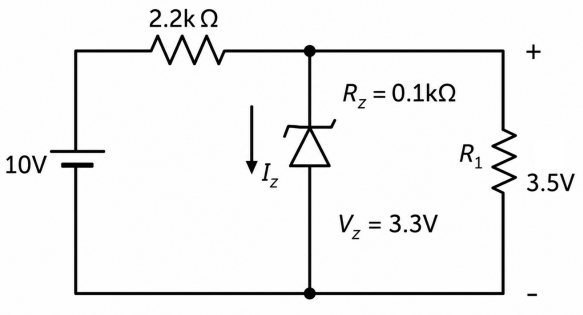


Fig. Q1.04.3

A. 33 mA **B.** 3.3 mA **C.** 2 mA **D.** 0 mA

Q1.03.1 MCQ 2003 2M

A voltage signal $10\sin\omega t$ is applied to the circuit with ideal diodes, as shown in Figure. The maximum and minimum values of the output waveform V_{out} of the circuit are respectively

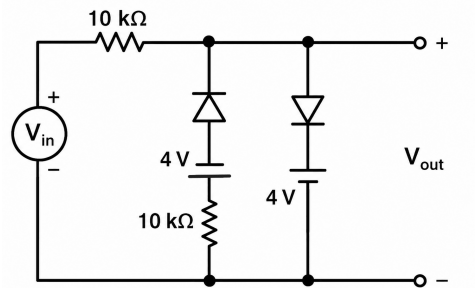


Fig. Q1.03.1

A. +10 V and −10 V **B.** +4 V and −4 V
C. +7 V and −4 V **D.** +4 V and −7 V

Q1.02.1 MCQ 2002 1 M

The cut-in voltage of both Zener diode D_z and diode D shown in Figure is 0.7 V, while break down voltage of D_z is 3.3 V and reverse breakdown voltage of D is 50V. The other parameters can be assumed to be the same as those of an ideal diode. The values of the peak output voltage (V_o) are

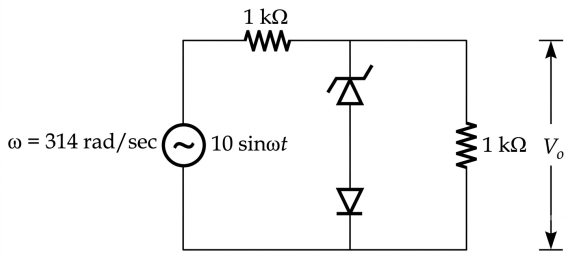


Fig. Q1.02.1

A. 3.3 V in the positive half cycle and 1.4 V in the negative half cycle

B. 4 V in the positive half cycle and 5 V in the negative half cycle

C. 3.3 V in both positive and negative half cycles

D. 4 V in both positive and negative half cycles

Q1.02.2 MCQ 2002 1 M

The forward resistance of the diode shown in Figure is 5 Ω and the remaining parameters are same as those of an ideal diode. The dc component of the source current is

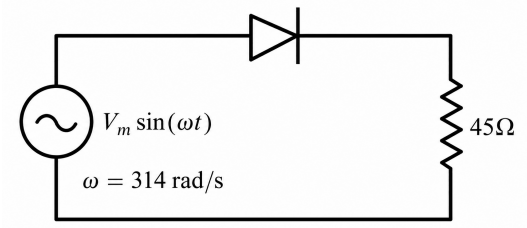


Fig. Q1.02.2

A. $\frac{V_m}{50\pi}$ **B.** $\frac{V_m}{50\pi\sqrt{2}}$ **C.** $\frac{V_m}{100\pi\sqrt{2}}$ **D.** $\frac{2V_m}{50\pi}$

Q1.00.1 MCQ 2000 2M

A diode whose terminal characteristics are related as $i_D = I_s e^{\left(\frac{v}{V_T}\right)}$, where I_s is the reverse saturation current and V_T is the thermal voltage (= 25mV), is biased at i_D = 2mA. Its dynamic resistance is:

A. 25 ohms **B.** 12.5 ohms **C.** 50 ohms **D.** 100 ohms

Q1.99.1 MCQ 1999 1M

As the temperature is increased, the voltage across a diode carrying a constant current

A. Increases

B. Decreases

C. Remains constant

D. May increase or decrease depending upon the doping levels in the junction

Q1.99.2 MCQ 1999 1M

The mobility of an electron in a conductor is expressed in terms of

A. $cm^2/V–s$ **B.** cm/V–s **C.** cm^2/V **D.** cm^2/s

Q1.96.1 MCQ 1996 1M

The depletion region or space charge region or transition in a semiconductor p-n junction diode has

A. Electrons and holes

B. Positive ions and electrons

C. Positive ions and negative ions

D. Negative ions and holes

E. No ions, electrons or holes

Q1.92.1 MCQ 1992 1M

In the following circuit the 5 V zener diode requires a minimum current of 10 mA. For obtaining a regulated output of 5 V, the maximum permissible load current (I_L) is _____mA and the minimum power rating of zener diode is _____W.

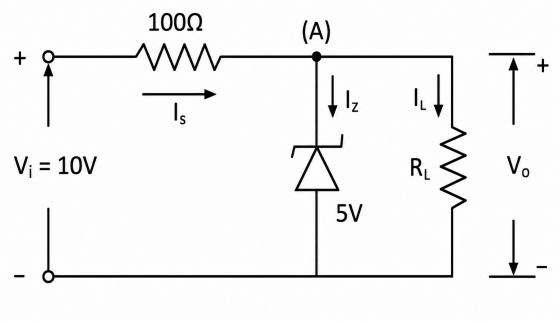


Fig. Q1.92.1

A. 40 mA and 0.25 W **B.** 30 mA and 0.20 W

C. 50 mA and 0.30 W **D.** 35 mA and 0.15 W

Q1.91.1 NAT 1991 2M

Figure shows an electric voltage regulator. The zener diode may be assumed to require a minimum current of 25 mA for satisfactory operations. The value of R required for satisfactory voltage regulation of the circuit is

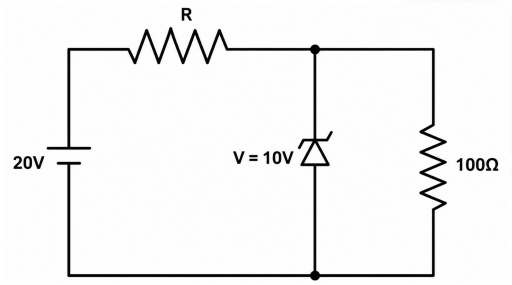


Fig. Q1.91.1

Q1.12.1 D	Q1.11.1 C	Q1.10.1 B
Q1.09.1 A	Q1.08.1 D	Q1.08.2 A
Q1.06.1 A	Q1.06.2 A	Q1.05.1 A
Q1.04.1 C	Q1.04.2 A	Q1.04.3 C
Q1.03.1 D	Q1.02.1 B	Q1.02.2 A
Q1.00.1 B	Q1.99.1 B	Q1.99.2 A
Q1.96.1 C	Q1.92.1 A	Q1.91.1 80

Answer Key

Q1.26.1 32.3 to 32.5	Q1.26.2 1.04 to 1.14	Q1.24.1 1.64 to 1.70
Q1.23.1 B,C	Q1.22.1 A	Q1.21.1 1.20 to 1.23
Q1.21.2 29.00 to 31.00	Q1.21.3 389 to 391	Q1.20.1 B
Q1.20.2 D	Q1.17.1 B	Q1.17.2 A
Q1.15.1 C	Q1.14.1 B	Q1.14.2 209 to 301
Q1.14.3 23 to 23	Q1.13.1 B	Q1.13.2 D

UNIT

II

MOSFETs

Topics

1. Introduction to MOS - Depletion and Enhancement Types, MOS Physics
2. Biasing a MOSFET - Transconductance, Output Impedance and Small Signal Analysis
3. Small Signal Model of MOSFET and Port Impedance
4. Common Source Amplifier - With and Without Degeneration, Different Biasing
5. Miller's Theorem and GmRout Method for Gain
6. Common Gate, Common Drain and PMOS Amplifiers
7. Cascode Current Source, Amplifiers and Current Mirror Circuits
8. Multistage Amplifiers and Pass Transistor Logic
9. Frequency Response of MOSFET and Common Gate Amplifiers
10. Introduction to Differential Amplifiers, Current Source, Voltage Source as Tail
11. Large Signal Analysis of Differential Amplifiers, Concept of Half Circuit
12. Building a Two-Stage Op Amp
13. Types and Effects of Feedback
14. Feedback Topologies - Voltage-Voltage, Voltage-Current, Current-Voltage, Current-Current
15. MOS Amplifiers with Body Effect

Unit Snapshot*

Stream: EE	**Questions:** 15	**MCQ:** 14	**MSQ:** 0	**NAT:** 1
1M: 8	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2022)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.25.1 MCQ 2025 1M

Which one of the following statements is true about the small signal voltage gain of a MOSFET based single stage amplifier?

A. Common source and common gate amplifiers are both inverting amplifiers

B. Common source and common gate amplifiers are both non-inverting amplifiers

C. Common source amplifier is inverting and common gate amplifier is non-inverting amplifier

D. Common source amplifiers is non-inverting and common gate amplifier is inverting amplifier

Q2.25.2 MCQ 2025 2M

In the circuit, I_{DC} is an ideal current source. The transistors M_1 and M_2 are assumed to be biased in saturation, wherein V_{in} is the input signal and V_{DC} is fixed DC voltage. Both transistors have a small signal resistance of r_{ds} and transconductance of g_m. The small signal output impedance of this circuit is

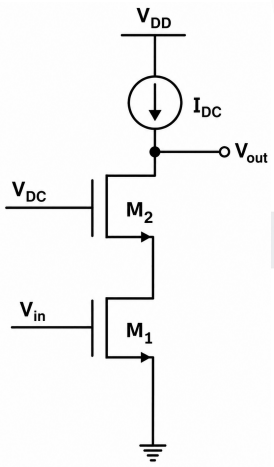


Fig. Q2.25.2

A. $2r_{ds}$

B. $\frac{1}{g_m} + r_{ds}$

C. $g_m r_{ds}^2 + 2r_{ds}$

D. infinity

Q2.22.1 MCQ 2022 1M

For an ideal MOSFET biased in saturation, the magnitude of the small signal current gain for a common drain amplifier is

A. 0 **B.** 1 **C.** 100 **D.** infinite

Q2.20.1 NAT 2020 1M

The cross-section of a metal-oxide-semiconductor structure is shown schematically. Starting from an uncharged condition, a bias of +3 V is applied to the gate contact with respect to the body contact. The charge inside the silicon dioxide layer is then measured to be $+Q$. The total charge contained within the dashed box shown, upon application of bias, expressed as a multiple of Q (absolute value in Coulombs, rounded off to the nearest integer) is _____.

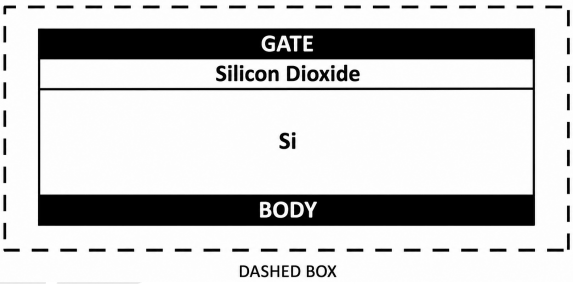


Fig. Q2.20.1

Q2.20.2 MCQ 2020 1M

A common-source amplifier with a drain resistance, R_D = 4.7 kΩ, is powered using a 10 V power supply. Assuming that the transconductance, g_m is 520 μA/V, the voltage gain of the amplifier is closest to:

A. 1.22 **B.** −1.22 **C.** 2.44 **D.** −2.44

Q2.19.1 MCQ 2019 2M

The enhancement type MOSFET in the circuit below operates according to the square law. $\mu_n C_{ox}$ = 100 μA/V^2, the threshold voltage (V_T) is 500 mV. Ignore channel length modulation. The output voltage V_{out} is

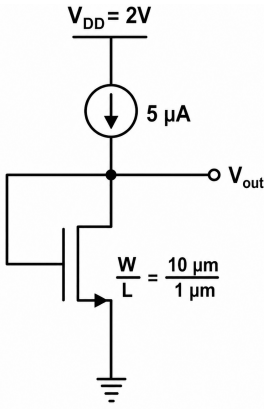


Fig. Q2.19.1

A. 500 mV **B.** 600 mV **C.** 2 V **D.** 100 mV

Q2.19.2 MCQ 2019 1M

Given V_{gs} is the gate-source voltage, V_{ds} is the drain source voltage, and V_{th} is the threshold voltage of an enhancement type NMOS transistor, the conditions for transistor to be biased in saturation are

A. $V_{gs} > V_{th}$; $V_{ds} \leq V_{gs} - V_{th}$

B. $V_{gs} < V_{th}$; $V_{ds} \leq V_{gs} - V_{th}$

C. $V_{gs} > V_{th}$; $V_{ds} \geq V_{gs} - V_{th}$

D. $V_{gs} < V_{th}$; $V_{ds} \geq V_{gs} - V_{th}$

Q2.05.1 MCQ 2005 1M

Assume that the N-channel $MOSFET$ shown in figure is ideal, and that its threshold voltage is 1 V. The voltage V_{ab} between nodes 'a' and 'b' is

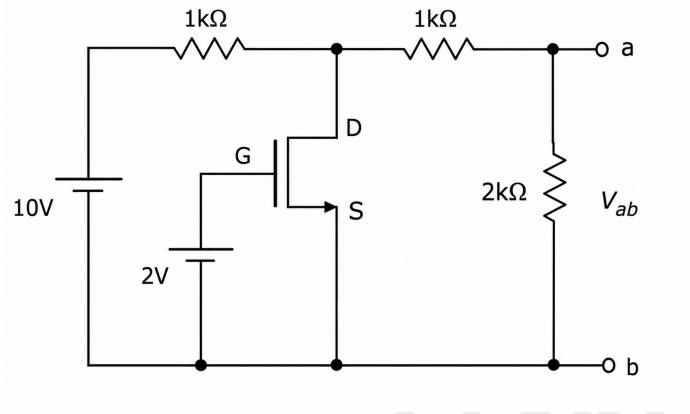


Fig. Q2.05.1

A. 5 V **B.** 2 V **C.** 1 V **D.** 0 V

Q2.05.2 MCQ 2005 2M

The charge distribution in a metal-dielectric-semiconductor specimen is shown in Figure. The negative charge density decreases linearly in the semiconductor as shown. The electric field distribution is as shown in

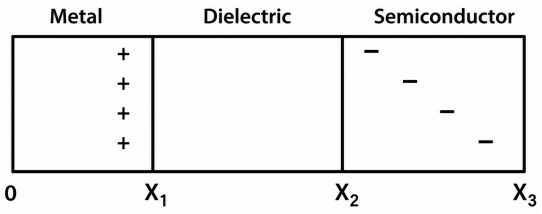
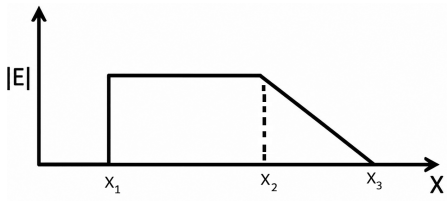
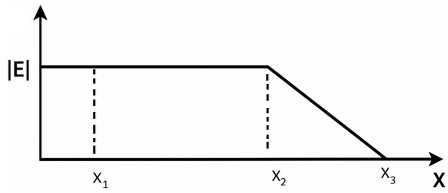


Fig. Q2.05.2

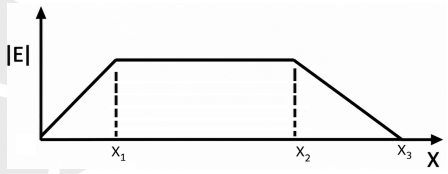
A.



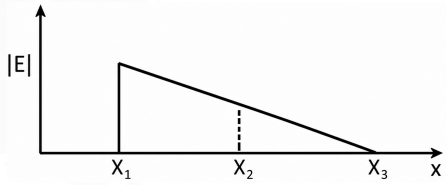
B.



C.



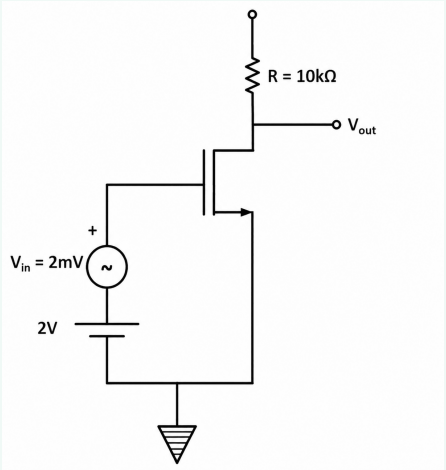
D.

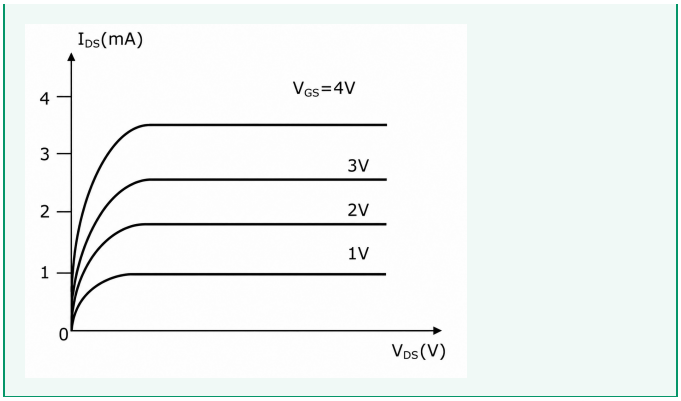


Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q2.05.3 and Q2.05.4 Assume that the threshold voltage of the N-channel MOSFET shown in figure is +0.75 V. The output characteristics of the MOSFET are also shown





Q2.05.3 MCQ 2005 2M

The transconductance of the MOSFET is

A. 0.75 ms **B.** 1 ms
C. 2 ms **D.** 10 ms

Q2.05.4 MCQ 2005 2M

The voltage gain of the amplifier is

A. +5 **B.** −7.5
C. +10 **D.** −10

Q2.04.1 MCQ 2004 2M

The value of R for which the $PMOS$ transistor in the figure, will be biased in linear region is

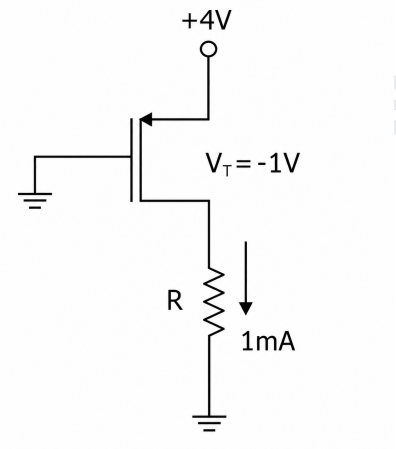


Fig. Q2.04.1

A. 220 Ω **B.** 470 Ω **C.** 680 Ω **D.** 1200 Ω

Q2.03.1 MCQ 2003 2M

For the n-channel enhancement $MOSFET$ shown in figure, the threshold voltage $V_{th} = 2$ V. The drain current I_D of the $MOSFET$ is 4 mA when the drain resistance R_D is 1 kΩ. If the value of R_D is increased to 4 kΩ, drain current I_D will become

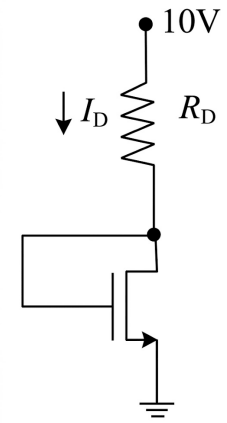


Fig. Q2.03.1

A. 2.8 mA **B.** 2.0 mA **C.** 1.4 mA **D.** 1.0 mA

Q2.03.2 MCQ 2003 1M

The variation of drain current with gate-to-source voltage ($I_D - V_{GS}$ characteristic) of a MOSFET is shown in Figure. The MOSFET is

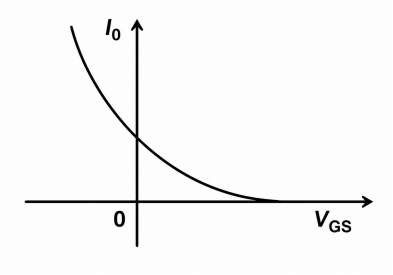


Fig. Q2.03.2

A. An n-channel depletion mode device

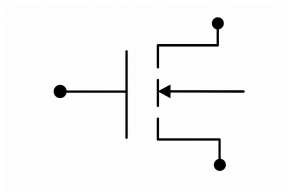
B. An n-channel enhancement mode device

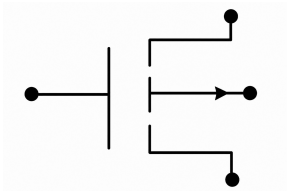
C. A p-channel depletion mode device

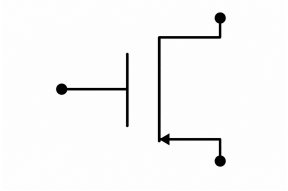
D. A p-channel enhancement mode device

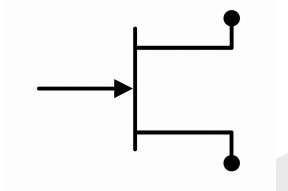
Q2.99.1 MCQ 1999 1M

An enhancement type *n*-channel MOSFET is represented by the symbol

A. 

B. 

C. 

D. 

Answer Key

Q2.25.1 C	Q2.25.2 C	Q2.22.1 D
Q2.20.1 0	Q2.20.2 D	Q2.19.1 B
Q2.19.2 C	Q2.05.1 D	Q2.05.2 A
Q2.05.3 B	Q2.05.4 D	Q2.04.1 D
Q2.03.1 C	Q2.03.2 C	Q2.99.1 A

UNIT

III

Operational Amplifier

Topics

1. Basics of Op-Amp, Comparator and Virtual Short
2. Inverting and Non-Inverting Amplifiers
3. Adder, Differential and Instrumentation Amplifiers
4. Log, Anti-Log and Rectifier Using Op-Amp
5. Filters Using Op-Amp
6. Offsets in Op-Amp
7. CMRR, Slew Rate and Frequency Response of Op-Amp
8. Integrator Using Op-Amp
9. Differentiator Using Op-Amp
10. Active Clipper, Clamper and Peak Detector
11. Multivibrators Using Op-Amp
12. Oscillators Using Op-Amp
13. Schmitt Triggers

Unit Snapshot*

Stream: EE	**Questions:** 81	**MCQ:** 53	**MSQ:** 0	**NAT:** 20
1M: 27	**2M:** 41	**Descriptive:** 13	**Avg Weightage** ~3M	**Range:** 1M(2018)–5M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

In the circuit shown, the open loop gain of the operational amplifier is $A_0 = 105$.

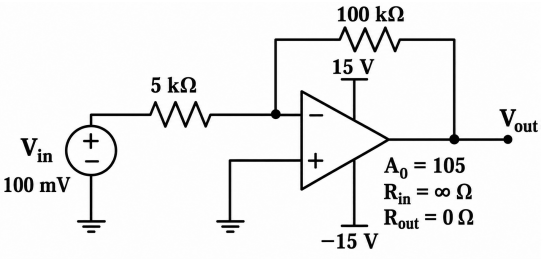


Fig. Q3.26.1

What is the voltage gain of the circuit?

A. -16.67 **B.** -20.00 **C.** -21.00 **D.** -12.67

Q3.25.1 MCQ 2025 1M

Assuming ideal op-amps, the circuit represents a

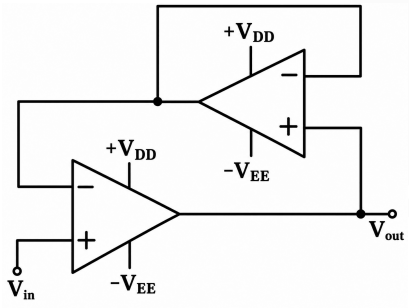


Fig. Q3.25.1

A. summing amplifier **B.** difference amplifier
C. logarithmic amplifier **D.** buffer

Q3.25.2 NAT 2025 2M

The op-amps in the following circuit are ideal. The voltage gain of the circuit is (round off to the nearest integer)

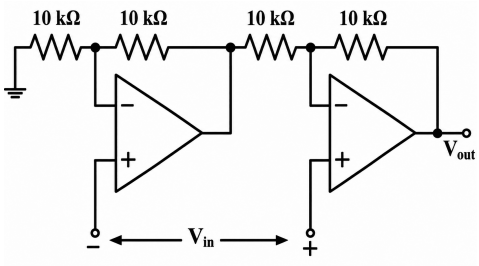


Fig. Q3.25.2

Q3.24.1 NAT 2024 2M

A difference amplifier is shown in the figure. Assume the op-amp to be ideal. The CMRR (in dB) of the difference amplifier is (rounded off to 2 decimal places)

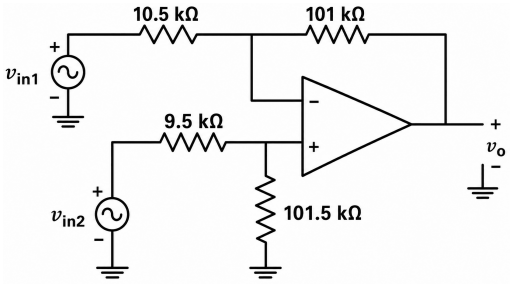


Fig. Q3.24.1

Q3.23.1 MCQ 2023 2M

Consider the OP AMP based circuit shown in the figure. Ignore the conduction drops of diodes D_1 and D_2. All the components are ideal and the breakdown voltage of the Zener is 5 V. Which of the following statements is true?

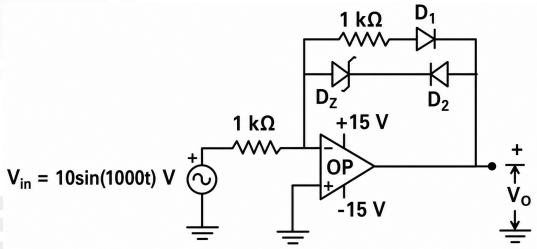


Fig. Q3.23.1

A. The maximum and minimum values of the output voltage V_O are +15 V and −10 V, respectively.

B. The maximum and minimum values of the output voltage V_O are +5 V and −15 V, respectively.

C. The maximum and minimum values of the output voltage V_O are +10 V and −5 V, respectively.

D. The maximum and minimum values of the output voltage V_O are +5 V and −10 V, respectively.

Q3.22.1 MCQ 2022 1M

The steady state output (V_{out}), of the circuit shown below, will

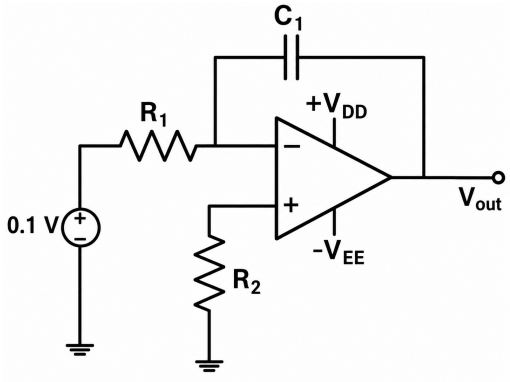


Fig. Q3.22.1

A. saturate to $+V_{DD}$ **B.** saturate to $-V_{EE}$
C. become equal to 0.1 V **D.** become equal to −0.1 V

Q3.22.2 MCQ 2022 2M

The output impedance of a non-ideal operational amplifier is denoted by Z_{out}. The variation in the magnitude of Z_{out} with increasing frequency, f, in the circuit shown below, is best represented by

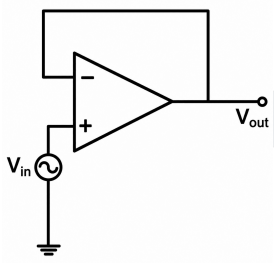
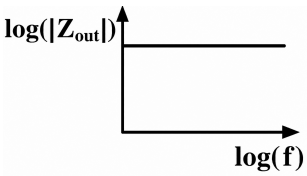
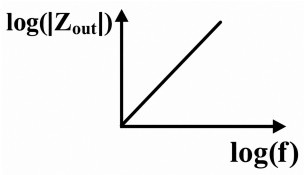
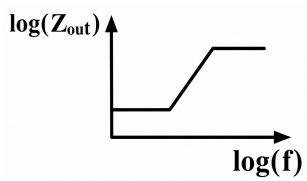
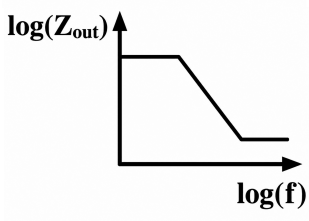


Fig. Q3.22.2

A.


B.


C.


D.


Q3.22.3 MCQ 2022 2M

The current gain (I_{out}/I_{in}) in the circuit with an ideal current amplifier given below is

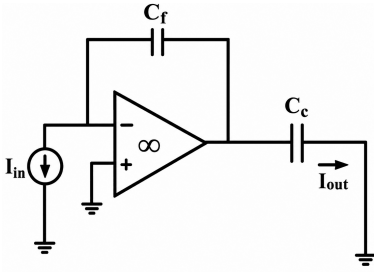


Fig. Q3.22.3

A. $\frac{C_f}{C_c}$ **B.** $\frac{-C_f}{C_c}$ **C.** $\frac{C_c}{C_f}$ **D.** $\frac{-C_c}{C_f}$

Q3.21.1 NAT 2021 2M

A CMOS Schmitt-trigger inverter has a low output level of 0 V and a high output level of 5 V. It has input thresholds of 1.6 V and 2.4 V. The input capacitance and output resistance of the Schmitt-trigger are negligible. The frequency of the oscillator shown is (Round off to 2 decimal places.)

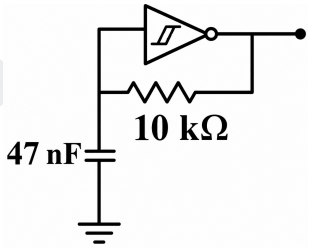


Fig. Q3.21.1

Q3.20.1 NAT 2020 2M

The temperature of the coolant oil bath for a transformer is monitored using the circuit shown. It contains a thermistor with a temperature-dependent resistance, $R_{thermistor} = 2\ (1 + \alpha T)$ kΩ, where T is the temperature in °C. The temperature coefficient, α, is $-(4 \pm 0.25)$ %/°C. Circuit parameters: $R_1 = 1$ kΩ, $R_2 = 1.3$ kΩ, $R_3 = 2.6$ kΩ. The error in the output signal (in V, rounded off to 2 decimal places) at 150°C is

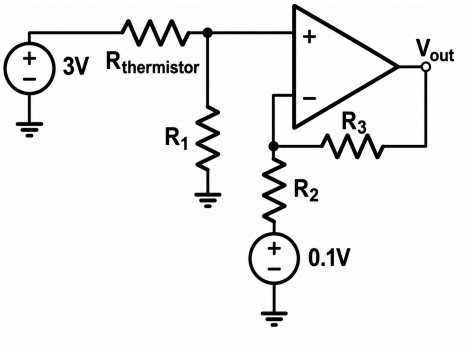


Fig. Q3.20.1

Q3.19.1 MCQ 2019 2M

In the circuit below, the operational amplifier is ideal. If $V_1 = 10\,\text{mV}$ and $V_2 = 50\,\text{mV}$, the output voltage (V_{out}) is

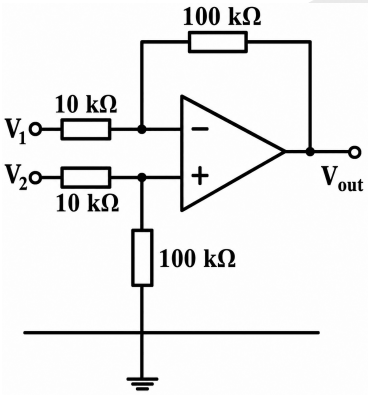


Fig. Q3.19.1

A. 100 mV **B.** 400 mV **C.** 500 mV **D.** 600 mV

Q3.18.1 MCQ 2018 1M

The op-amp shown in the figure is ideal. The input impedance $\frac{v_{in}}{i_{in}}$ is given by

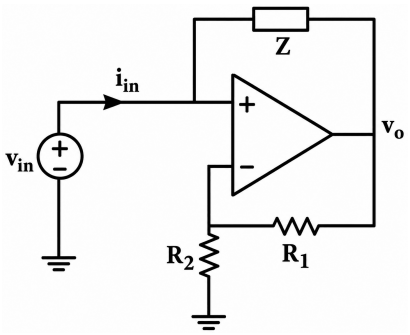


Fig. Q3.18.1

A. $Z\frac{R_1}{R_2}$ **B.** $-Z\frac{R_2}{R_1}$ **C.** Z **D.** $-Z\frac{R_1}{R_1+R_2}$

Q3.17.1 MCQ 2017 2M

The approximate transfer characteristic for the circuit shown below with an ideal operational amplifier and diode will be

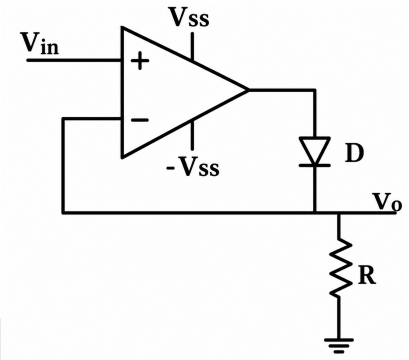
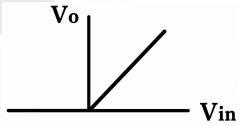
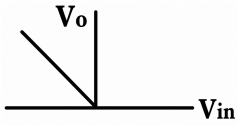
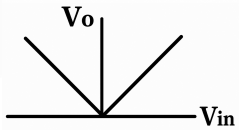
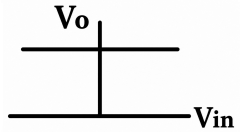


Fig. Q3.17.1

A.


B.


C.


D.


Q3.17.2 MCQ 2017 2M

For the circuit shown below, assume that the OPAMP is ideal.

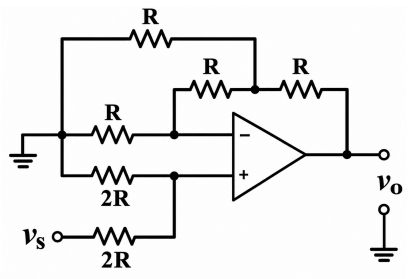


Fig. Q3.17.2

Which one of the following is TRUE?

A. $v_O = v_S$ **B.** $v_O = 1.5\,v_S$
C. $v_O = 2.5\,v_S$ **D.** $v_O = 5\,v_S$

Q3.16.1 MCQ 2016 1M

The circuit shown below is an example of a

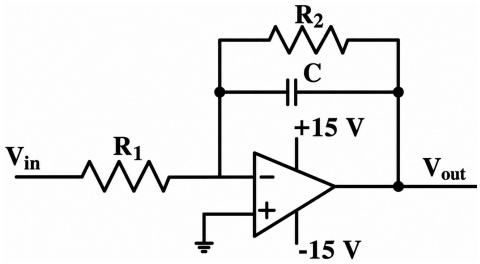


Fig. Q3.16.1

A. low pass filter
B. band pass filter
C. high pass filter
D. notch filter

Q3.16.2 MCQ 2016 2M

For the circuit shown below, taking the opamp as ideal, the output voltage V_{out} in terms of the input voltages V_1, V_2 and V_3 is

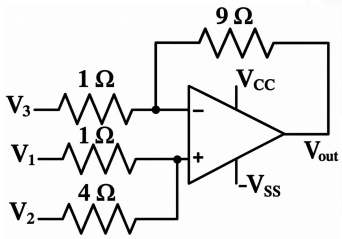


Fig. Q3.16.2

A. $1.8V_1 + 7.2V_2 - V_3$
B. $2V_1 + 8V_2 - 9V_3$
C. $7.2V_1 + 1.8V_2 - V_3$
D. $8V_1 + 2V_2 - 9V_3$

Q3.15.1 MCQ 2015 1M

Of the four characteristics given below, which are the major requirements for an instrumentation amplifier?

P. High common mode rejection ratio

Q. High input impedance

R. High linearity

S. High output impedance

A. P, Q and R only
B. P and R only
C. P, Q and S only
D. Q, R and S only

Q3.15.2 MCQ 2015 1M

Consider the circuit shown in the figure. In this circuit $R = 1\ \text{k}\Omega$, and $C = 1\ \mu\text{F}$. The input voltage is sinusoidal with a frequency of 50 Hz, represented as a phasor with magnitude V_i and phase angle 0 radian as shown in the figure. The output voltage is represented as a phasor with magnitude V_o and phase angle δ radian. What is the value of the output phase angle δ (in radian) relative to the phase angle of the input voltage?

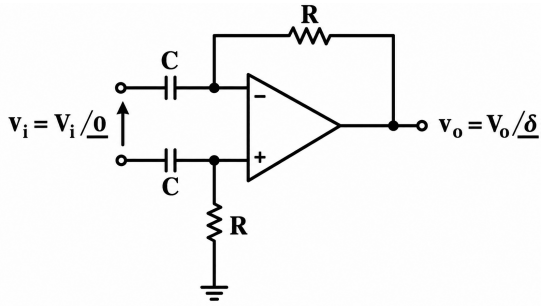


Fig. Q3.15.2

A. 0
B. π
C. $\pi/2$
D. $-\pi/2$

Q3.15.3 NAT 2015 1M

The operational amplifier shown in the figure is ideal. The input voltage (in Volt) is $V_i = 2\sin(2\pi \times 2000t)$. The amplitude of the output voltage V_o (in Volt) is

_____

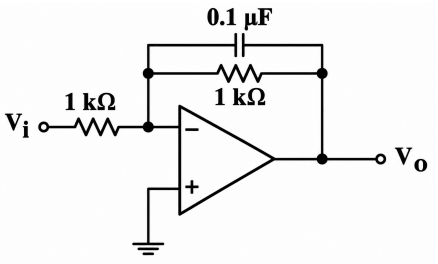


Fig. Q3.15.3

Q3.15.4 MCQ 2015 2M

The op-amp shown in the figure has a finite gain $A = 1000$ and an infinite input resistance. A step-voltage $V_i = 1$ mV is applied at the input at time $t = 0$ as shown. Assuming that the operational amplifier is not saturated, the time constant (in millisecond) of the output voltage V_o is

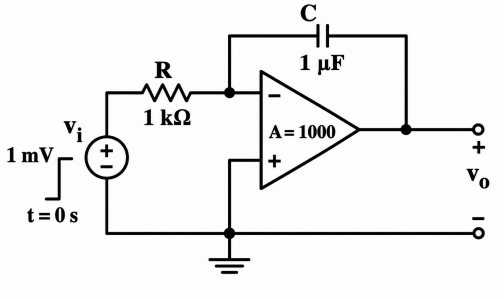


Fig. Q3.15.4

A. 1001 **B.** 101 **C.** 11 **D.** 1

Q3.15.5 MCQ 2015 2M

The saturation voltage of the ideal op-amp shown below is ±10 V. The output voltage v_0 of the following circuit in the steady-state is

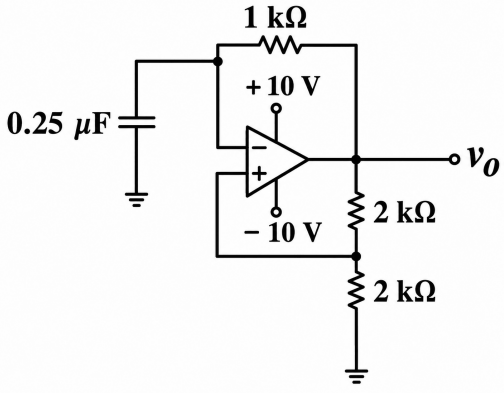


Fig. Q3.15.5

A. square wave of period 0.55 ms.

B. triangular wave of period 0.55 ms.

C. square wave of period 0.25 ms.

D. triangular wave of period 0.25 ms.

Q3.15.6 MCQ 2015 2M

The filters $F1$ and $F2$ having characteristics as shown in Fig. (a) and (b) are connected as shown in Fig. (c).

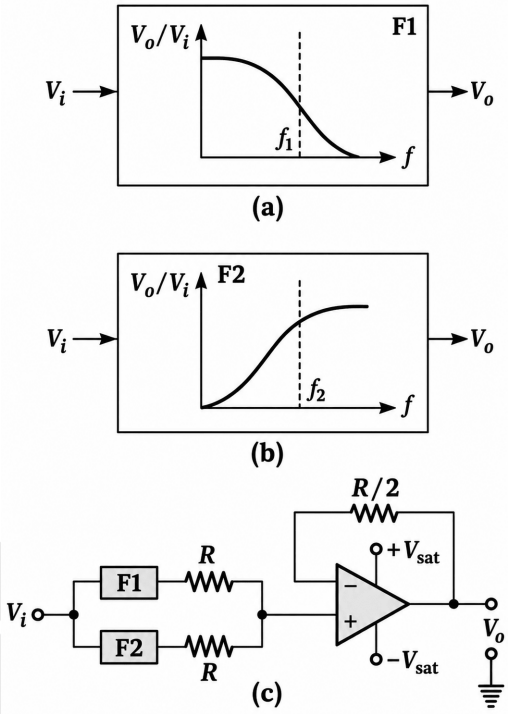


Fig. Q3.15.6

The cut-off frequencies of $F1$ and $F2$ are f_1 and f_2 respectively. If $f_1 < f_2$, the resultant circuit exhibits the characteristic of a

A. Band-pass filter **B.** Band-stop filter

C. All-pass filter **D.** High Q-filter

Q3.14.1 MCQ 2014 1M

An operational-amplifier circuit is shown in the figure.

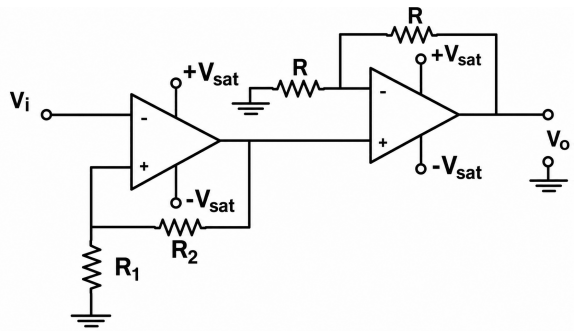


Fig. Q3.14.1

The output of the circuit for a given input v_i is

A. $-\left(\frac{R_2}{R_1}\right) v_i$ **B.** $-\left(1+\frac{R_2}{R_1}\right) v_i$

C. $\left(1+\frac{R_2}{R_1}\right) v_i$ **D.** $+V_{sat}$ or $-V_{sat}$

Q3.14.2 MCQ 2014 2M

Given that the op-amps in the figure are ideal, the output voltage V_0 is

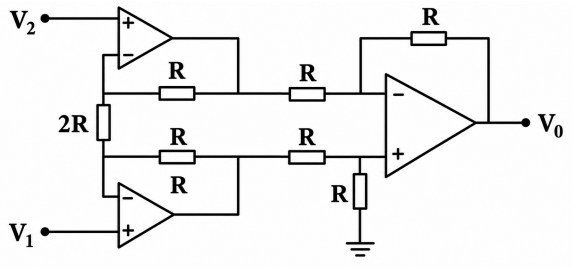


Fig. Q3.14.2

A. $(V_1 - V_2)$
B. $2(V_1 - V_2)$
C. $(V_1 - V_2)/2$
D. $(V_1 + V_2)$

Q3.14.3 NAT 2014 2M

An oscillator circuit using ideal op-amp and diodes is shown in the figure.

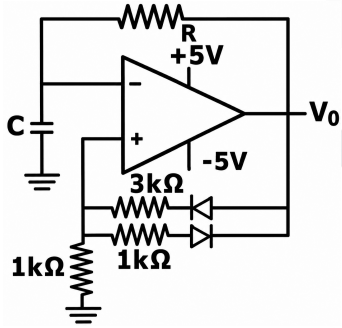


Fig. Q3.14.3

The time duration for +ve part of the cycle is Δt_1 and for −ve part is Δt_2. The value of $e^{\frac{\Delta t_1 - \Delta t_2}{RC}}$ will be

---

Q3.14.4 MCQ 2014 2M

The transfer characteristic of the Op-amp circuit shown in figure is

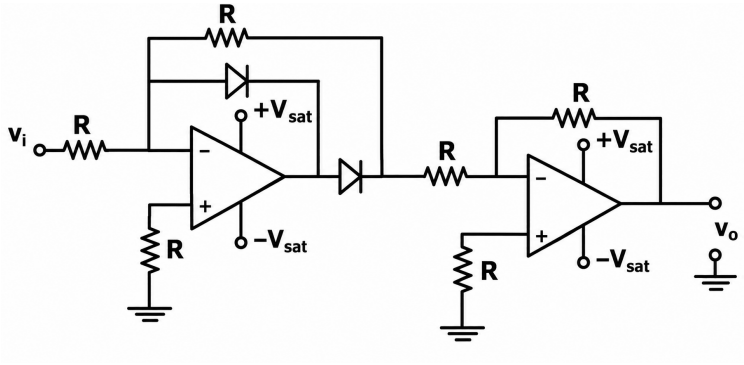
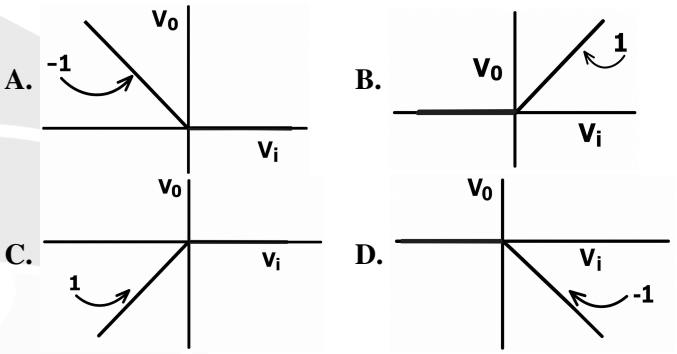


Fig. Q3.14.4



Q3.13.1 MCQ 2013 1M

In the circuit shown below what is the output voltage (V_{out}) in Volts if a silicon transistor Q and an ideal op-amp are used?

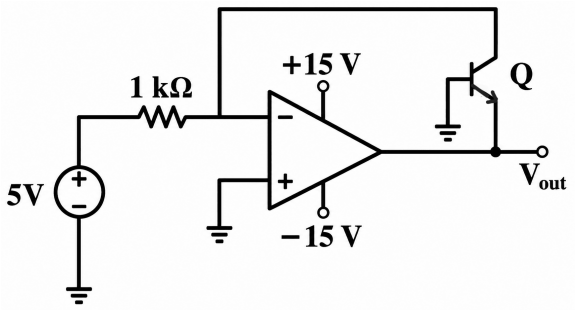


Fig. Q3.13.1

A. −15 **B.** −0.7 **C.** +0.7 **D.** +15

Q3.13.2 MCQ 2013 2M

In the circuit shown below the op-amps are ideal. Then V_{out} in Volts is

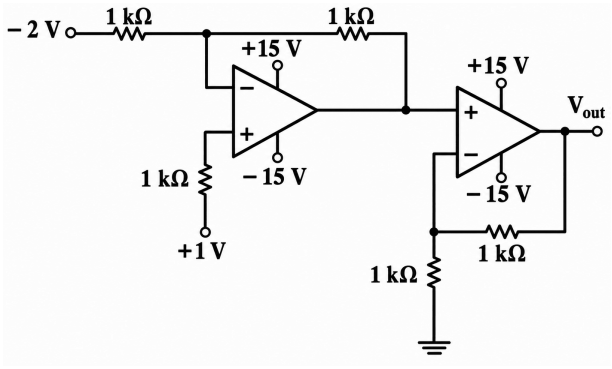


Fig. Q3.13.2

A. 4 **B.** 6 **C.** 8 **D.** 10

Q3.12.1 MCQ 2012 2M

The circuit shown is a

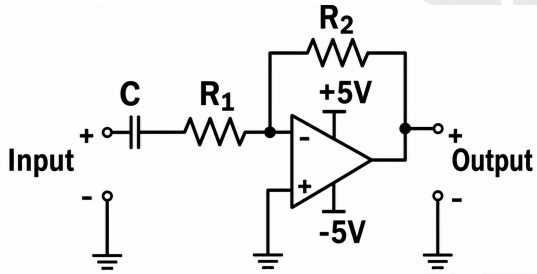


Fig. Q3.12.1

A. low pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

B. high pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

C. low pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

D. high pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

Q3.11.1 MCQ 2011 1M

A low-pass filter with a cut-off frequency of 30 Hz is cascaded with a high-pass filter with a cut-off frequency of 20 Hz. The resultant system of filters will function as

A. an all-pass filter **B.** an all-stop filter
C. a band stop (band-reject) filter **D.** a band-pass filter

Q3.11.2 MCQ 2011 1M

For the circuit shown below,

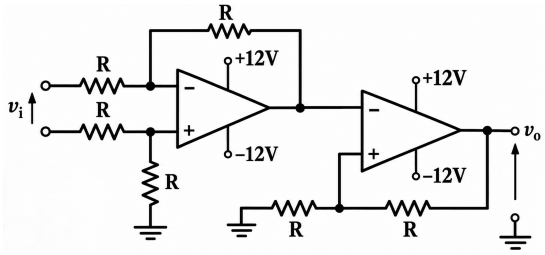
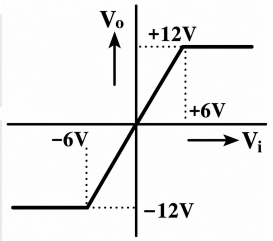


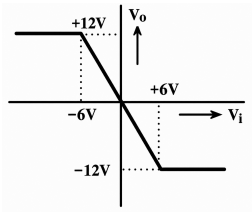
Fig. Q3.11.2

the CORRECT transfer characteristic is

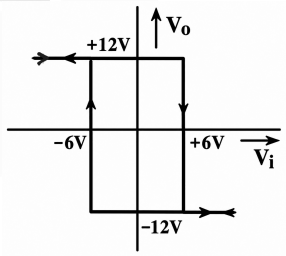
A.



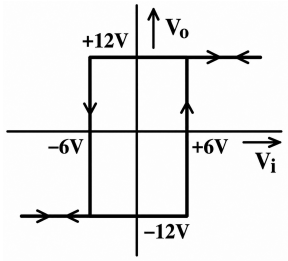
B.



C.



D.



Q3.10.1 MCQ 2010 1M

Given that the op-amp is ideal, the output voltage V_o is

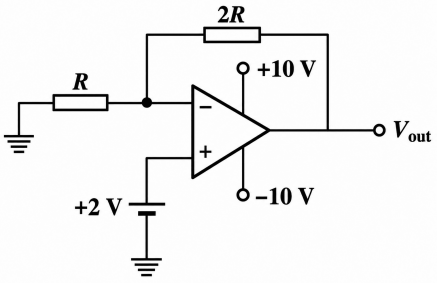


Fig. Q3.10.1

A. 4 V **B.** 6 V **C.** 7.5 V **D.** 12.12 V

Q3.09.1 MCQ 2009 1M

The nature of feedback in the op-amp circuit shown is

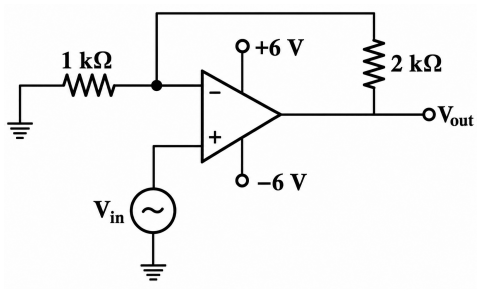


Fig. Q3.09.1

A. current-current feedback

B. voltage-voltage feedback

C. current-voltage feedback

D. voltage-current feedback

Q3.09.2 MCQ 2009 2M

The following circuit has $R = 10\,\text{k}\Omega$, $C = 10\,\mu\text{F}$. The input voltage is a sinusoid at 50Hz with an rms value of 10V. Under ideal conditions, the current i_S from the source is

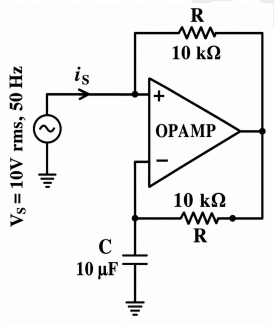


Fig. Q3.09.2

A. 10π mA leading by 90° **B.** 20π mA leading by 90°
C. 10 mA leading by 90° **D.** 10π mA lagging by 90°

Q3.09.3 MCQ 2009 2M

An ideal opamp circuit and its input waveform are shown in the figures. The output waveform of this circuit will be

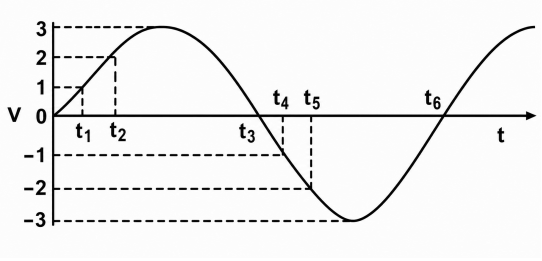


Fig. Q3.09.3

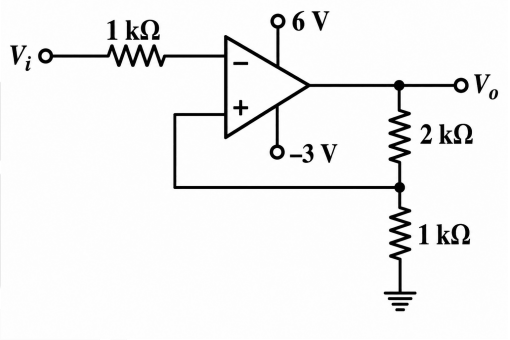
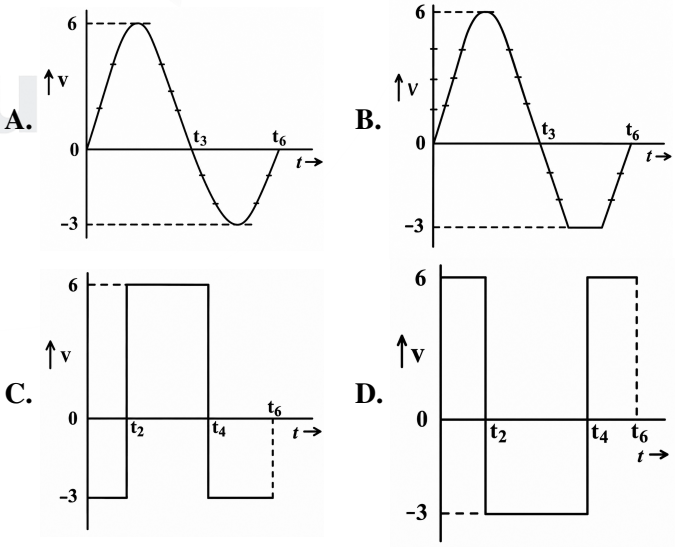


Fig. Q3.09.3



Q3.08.1 MCQ 2008 2M

The block diagrams of two of half wave rectifiers are shown in the figure with the transfer characteristics of the rectifiers.

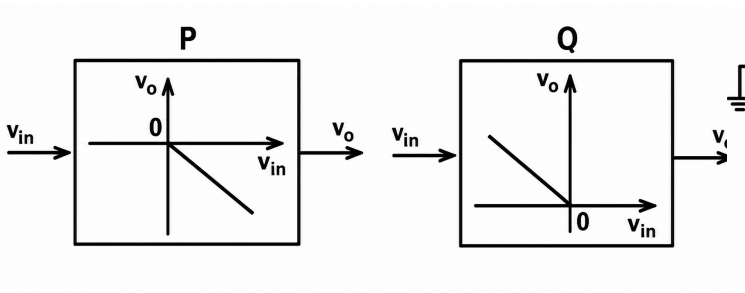


Fig. Q3.08.1

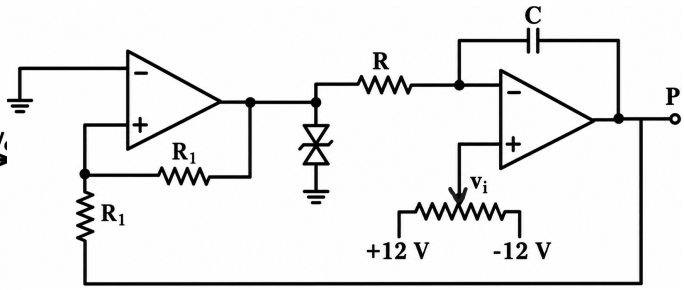
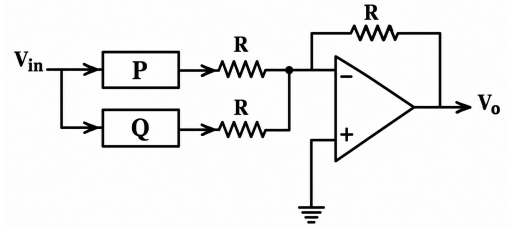
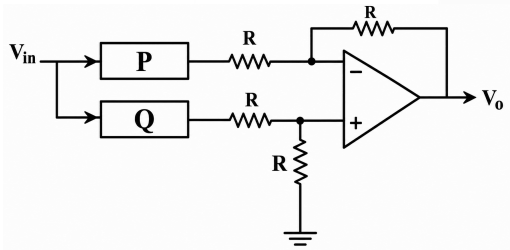
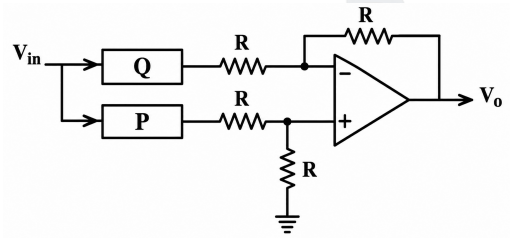


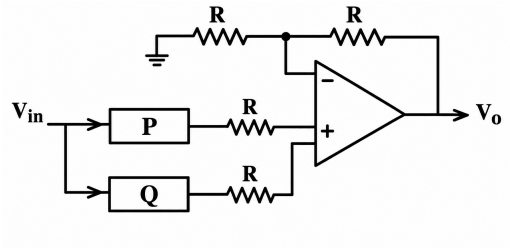
Fig. Q3.08.2

It is desired to make full wave rectifiers using above two-half-wave rectifiers. The resultants circuit will be

A.


B.


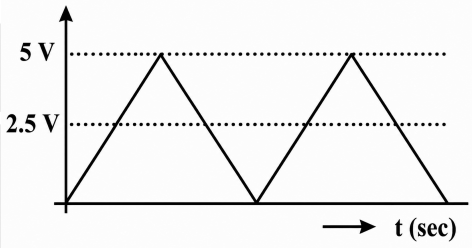
C.


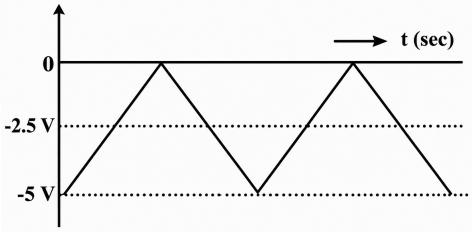
D.


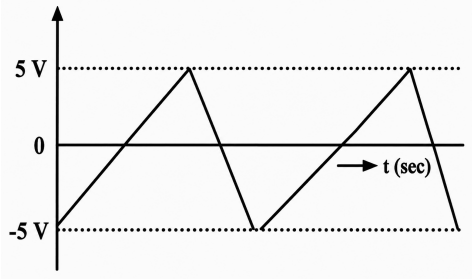
Q3.08.2 MCQ 2008 2M

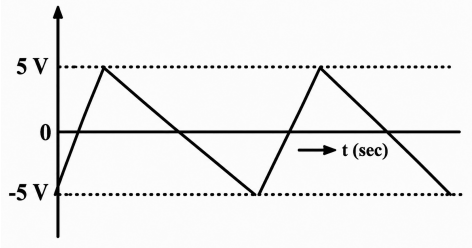
A waveform generator circuit using OPAMPs is shown in the figure. It produces a triangular wave at point 'P' with a peak to peak voltage of 5 V for v_i = 0 V.

If the voltage v_i is made +2.5V, the voltage waveform at point 'P' will become

A.


B.


C.


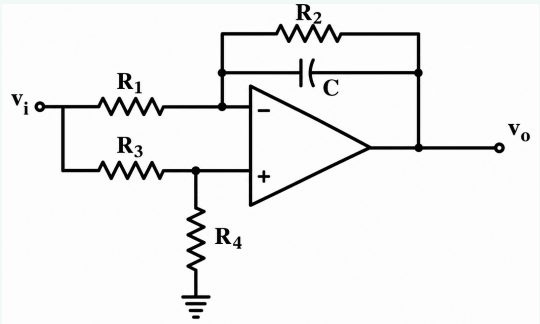
D.


Common Data Questions

The next 2 questions share the data below.

Commmon data for Q3.08.3 and Q3.08.4.

A general filter circuit is shown in the figure:



Q3.08.3 MCQ 2008 2M

If $R_1 = R_2 = R_A$ and $R_3 = R_4 = R_B$, the circuit acts as a

A. all pass filter **B.** band pass filter

C. high pass filter **D.** low pass filter

Q3.08.4 MCQ 2008 2M

The output of the filter in Q3.08.3 is given to the circuit shown in figure:

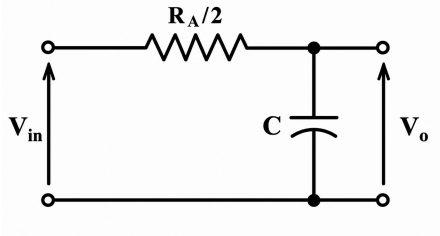
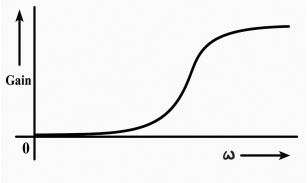


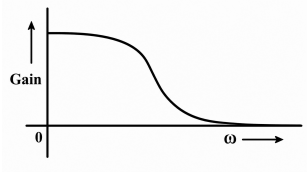
Fig. Q3.08.4

The gain vs frequency characteristic of the output (v_o) will be

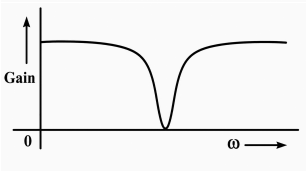
A.



B.



C.



D.

Q3.07.1 MCQ 2007 1M

The circuit shown in the figure is

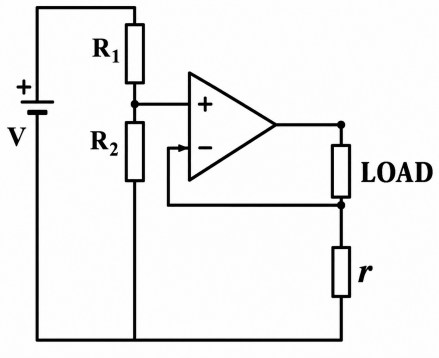


Fig. Q3.07.1

A. a voltage source with voltage $\frac{rV}{R_1 \parallel R_2}$

B. a voltage source with voltage $\frac{r \parallel R_2}{R_1} V$

C. a current source with current $\frac{r \parallel R_2}{R_1 + R_2} \cdot \frac{V}{r}$

D. a current source with current $\frac{R_2}{R_1 + R_2} \cdot \frac{V}{r}$

Q3.07.2 MCQ 2007 2M

The switch S in the circuit of the figure is initially closed. It is opened at time $t = 0$. You may neglect the Zener diode forward voltage drops. What is the behaviour of V_{OUT} for $t > 0$?

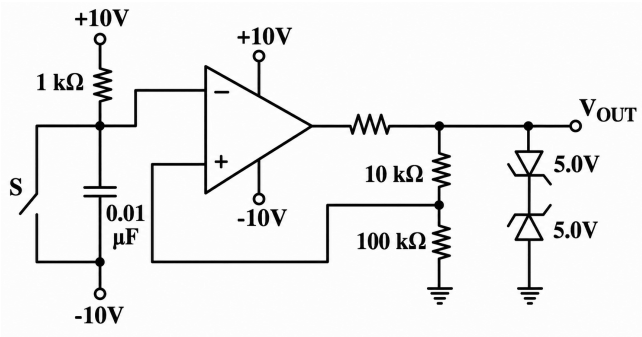


Fig. Q3.07.2

A. It makes a transition from −5 V to +5 V at $t = 12.98\ \mu s$

B. It makes a transition from −5 V to +5 V at $t = 2.57\ \mu s$

C. It makes a transition from +5 V to −5 V at $t = 12.98\ \mu s$

D. It makes a transition from +5 V to −5 V at $t = 2.57\ \mu s$

Q3.06.1 MCQ 2006 1M

For a given sinusoidal input voltage, the voltage waveform at point P of the clamper circuit shown in figure will be

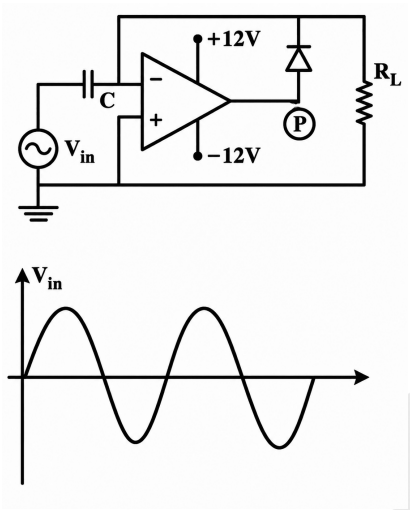
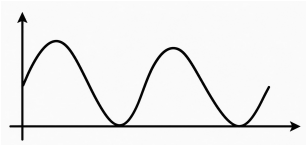
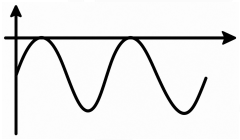
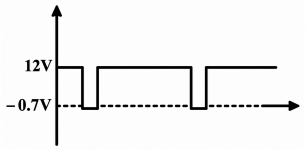
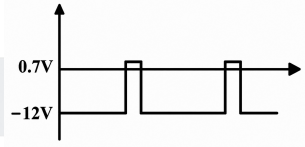


Fig. Q3.06.1

A. 

B. 

C. 

D. 

Q3.06.2 MCQ 2006 2M

A relaxation oscillator is made using OPAMP as shown in figure. The supply voltages of the OPAMP are ±12V. The voltage waveform at point P will be

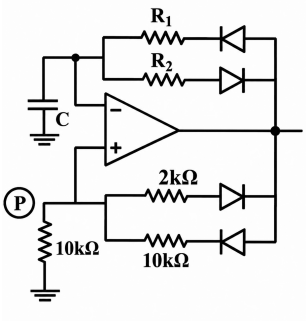
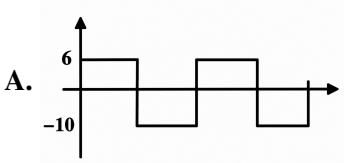
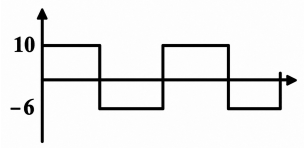
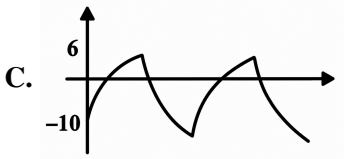
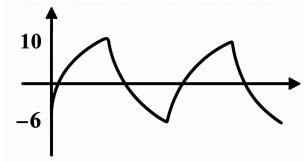


Fig. Q3.06.2

A. 

B. 

C. 

D. 

Q3.05.1 MCQ 2005 2M

Consider the inverting amplifier, using an ideal operational amplifier shown in Figure. The designer wishes to realize the input resistance seen by the small signal source to be as large as possible, while keeping the voltage gain between -10 and -25, the upper limit on R_F is 1 MΩ. The value of R_1 should be

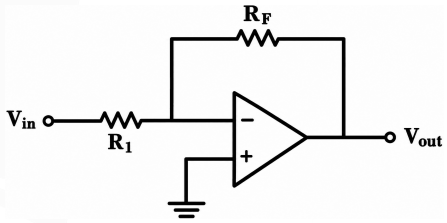


Fig. Q3.05.1

A. Infinity **B.** 1 MΩ **C.** 100 MΩ **D.** 40 kΩ

Q3.05.2 MCQ 2005 2M

In the given figure, if the input is a sinusoidal signal, the output will appear as shown in

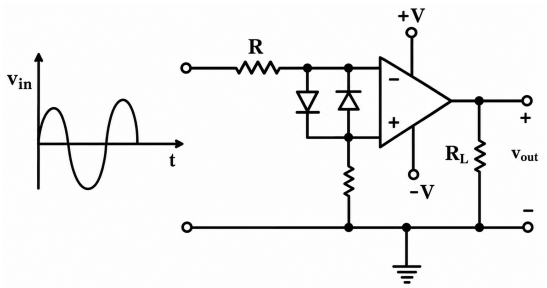
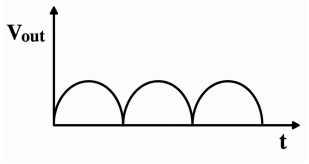
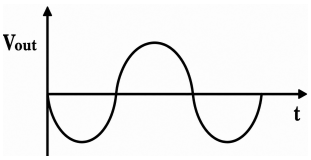
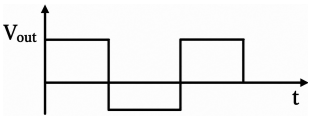
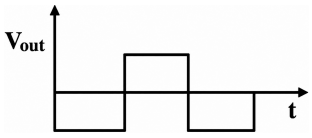


Fig. Q3.05.2

A.


B.


C.


D.


Q3.04.1 MCQ 2004 2M

The input resistance $R_{IN} = \left(\frac{v_x}{i_x}\right)$ of the circuit in figure is

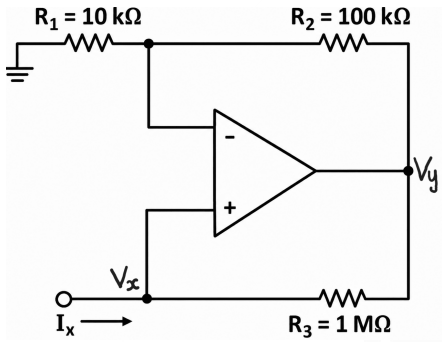


Fig. Q3.04.1

A. +100 kΩ **B.** −100 kΩ **C.** +1 MΩ **D.** −1 MΩ

Q3.04.2 MCQ 2004 2M

In the active filter circuit shown in figure, if $Q = 1$, a pair of poles will be realized with ω_0 equal to

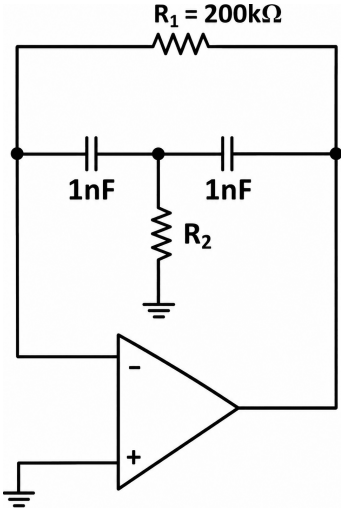


Fig. Q3.04.2

A. 1000 rad/s **B.** 100 rad/s **C.** 10 rad/s **D.** 1 rad/s

Q3.03.1 MCQ 2003 1M

For the circuit of Figure with an ideal operational amplifier, the maximum phase shift of the output V_{out} with reference to the input V_{in} is

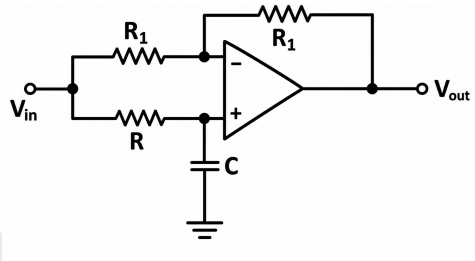


Fig. Q3.03.1

A. 0° **B.** −90° **C.** +90° **D.** ±180°

Q3.03.2 MCQ 2003 2M

Assuming the operational amplifier to be ideal, the gain V_{out}/V_{in} for the circuit shown in figure is

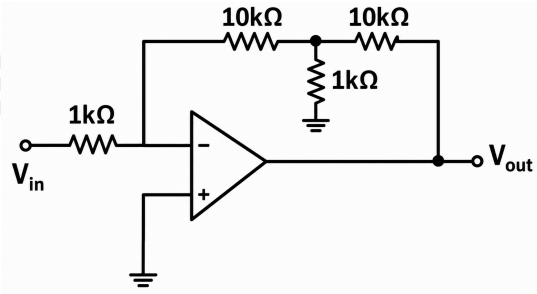


Fig. Q3.03.2

A. −1 **B.** −20 **C.** −100 **D.** −120

Q3.02.1 MCQ 2002 2M

The output voltage (v_o) of the Schmitt trigger shown in Figure swings between +15V and −15V. Assume that the operational amplifier is ideal. The output will change from +15V to −15V when the instantaneous value of the input sine wave is

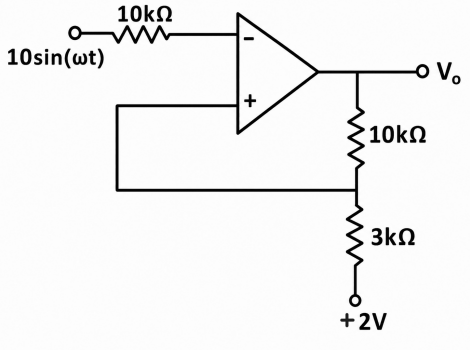


Fig. Q3.02.1

A. 5 V in the positive slope only

B. 5 V in the negative slope only

C. 5 V in the positive and negative slopes

D. 3 V in the positive and negative slopes

Q3.02.2 Descriptive 2002 5M

Determine the transfer function $\left(\frac{V_y}{V_x}\right)$ for the RC network shown in Fig.(a).

This network is used as a feedback circuit in an oscillator circuit shown in Fig.(b) to generate sinusoidal oscillations. Assuming that the operational amplifier is ideal, determine R_F for generating these oscillations. Also, determine the oscillation frequency if $R = 10$ kΩ and $C = 100$ pF.

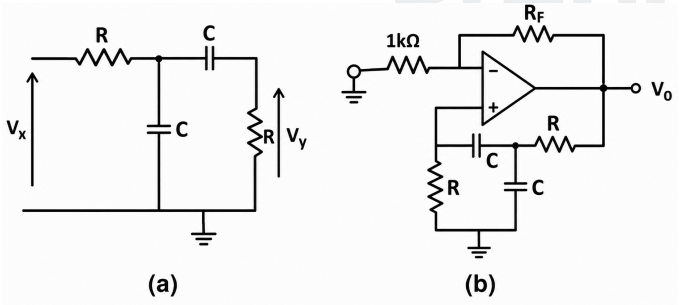


Fig. Q3.02.2

Q3.01.1 MCQ 2001 1M

An op-amp has an open-loop gain of 10^5 and an open-loop upper cutoff frequency of 10 Hz. If this op-amp is connected as an amplifier with a closed gain of 100, then the new upper cutoff frequency is

A. 10 Hz **B.** 100 Hz **C.** 10 kHz **D.** 100 kHz

Q3.01.2 MCQ 2001 2M

For the oscillator circuit shown in figure, the expression for the time period of oscillation can be given by (where $\tau = RC$)

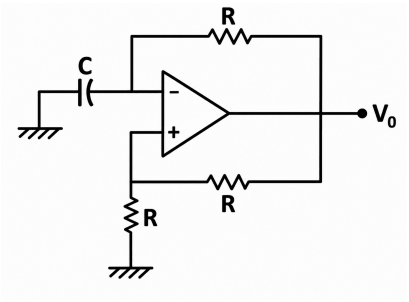


Fig. Q3.01.2

A. $\tau \ln 3$ **B.** $2\tau \ln 3$ **C.** $\tau \ln 2$ **D.** $2\tau \ln 2$

Q3.01.3 MCQ 2001 2M

An op-amp having a slew rate of 62.8 V/μsec is connected in a voltage follower configuration. If the maximum amplitude of the input sinusoidal is 10 Volts, then the minimum frequency at which the slew rate limited distortion would set in the output is

A. 1 MHz **B.** 6.28 MHz **C.** 10 MHz **D.** 62.8 MHz

Q3.01.4 NAT 2001 5M

For the op-amp circuit shown in figure, determine the output voltage V_0. Assume that the op-amps are ideal.

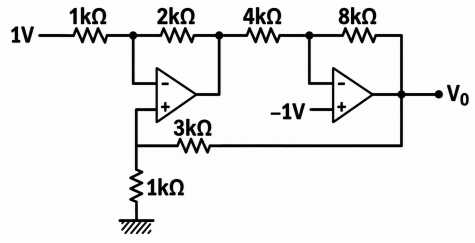


Fig. Q3.01.4

Q3.01.5 NAT 2001 5M

A simple active filter is shown in Figure. Assume ideal op-amp. Derive the transfer function $\frac{V_0}{V_i}$ of the circuit, and state the type of the filter (i.e., high-pass, low-pass, band-pass, or band-reject). Determine the required values of R_1,

R_2 and C in order for the filter to have a 3-dB frequency of 1 kHz, a high frequency input resistance of 100 kΩ, and a high frequency gain magnitude of 10.

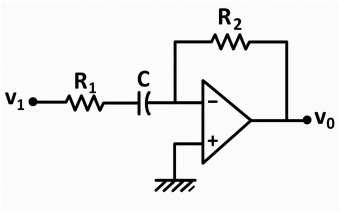


Fig. Q3.01.5

Q3.00.1 MCQ 2000 1M

The feedback factor for the circuit shown in figure is

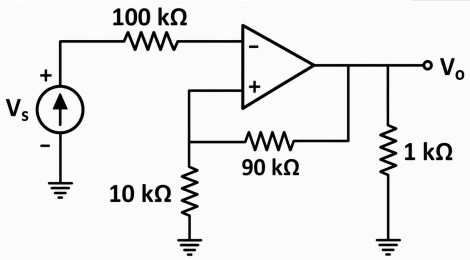


Fig. Q3.00.1

A. $\frac{9}{100}$ **B.** $\frac{9}{10}$ **C.** $\frac{1}{9}$ **D.** $\frac{1}{10}$

Q3.00.2 MCQ 2000 1M

The circuit shown in figure uses an ideal op-amp working with +5 V and −5 V power supplies. The output voltage V_o is equal to

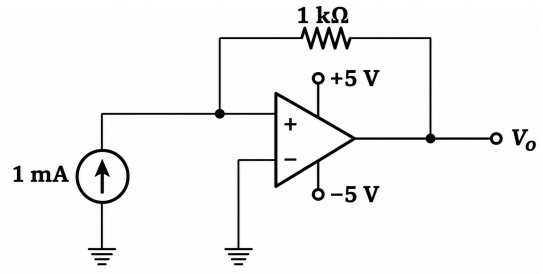


Fig. Q3.00.2

A. +5 V **B.** −5 V **C.** +1 V **D.** −1 V

Q3.00.3 NAT 2000 5M

An active filter consisting of an op-amp, resistors R_1, R_2, R_3 and two capacitors of value C each, has a transfer function

$$T(s) = \frac{\frac{-s}{(R_1C)}}{s^2 + \frac{2s}{(R_3C)} + \frac{1}{(RR_3C^2)}}$$

Where $R = R_1 \parallel R_2$

If $R_1 = 2$ kΩ, $R_2 = \frac{2}{3}$ kΩ, $R_3 = 200$ kΩ and $C = 0.1\,\mu$F, determine the center frequency ω_0, gain A_0 and the Q of the filter.

Q3.99.1 NAT 1999 5M

The input voltage V_i in the circuit shown in fig is a 1 kHz sine wave of 1 V amplitude. Assume ideal operational amplifiers with ±15 V DC supply. Sketch on a single diagram the waveforms of the voltages V_i and V_1 shown, indicating the peak value of V_1 and the average value V_0

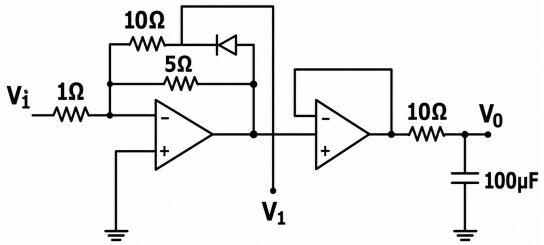


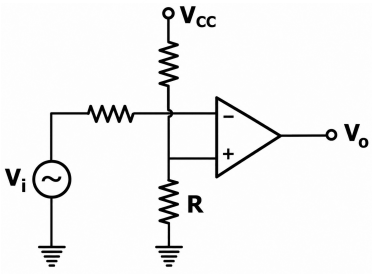
Fig. Q3.99.1

Q3.98.1 Descriptive 1998 2M

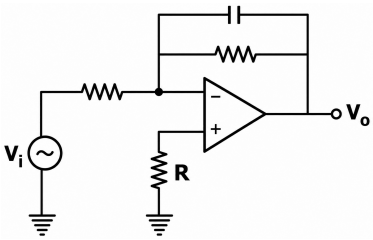
Match the following:

Circuits:

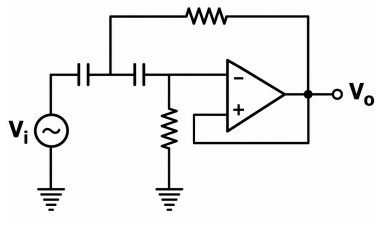
A.



B.



C.



Functions:

A. P. High-pass filter

B. Q. Amplifier

C. R. Comparator

D. S. Low-pass filter

Q3.98.2 Descriptive | 1998 | 5M

In the circuit of fig. Q3.98.2, $R_s = 2\ \text{k}\Omega$ and $R = 5\ \Omega$. For the op-amp, $A = 10^5$, $R_i = 100\ \text{k}\Omega$, and $R_0 = 50\ \Omega$.

For $V_0 = 10$ V, calculate V_s and $\frac{V_0}{V_s}$ and estimate the input resistance of the circuit.

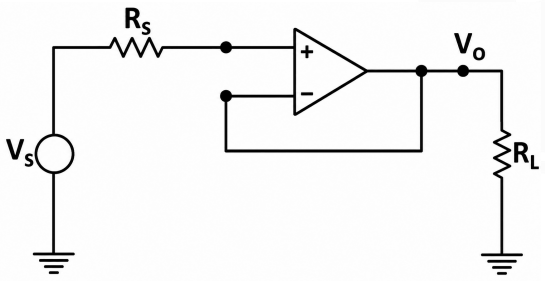


Fig. Q3.98.2

Q3.98.3 NAT | 1998 | 5M

Show that the circuit given in fig. will work as an oscillator at $f = \frac{1}{2\pi RC}$, if $R_1 = 2R_2$

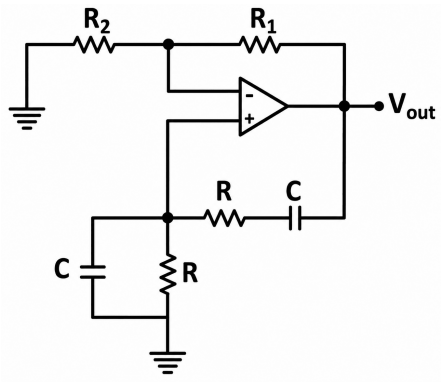


Fig. Q3.98.3

Q3.97.1 MCQ | 1997 | 1M

A major advantage of active filters is that they can be realized without using

A. op-amps **B.** inductors
C. resistors **D.** capacitors

Q3.97.2 Descriptive | 1997 | 2M

The circuit shown in figure, acts as a..... and for the given inputs, its output voltage is

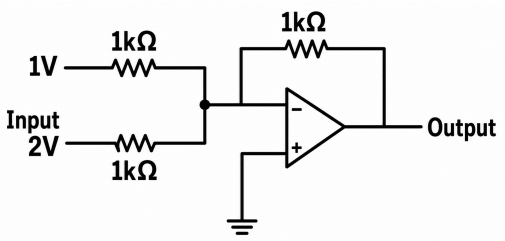


Fig. Q3.97.2

Q3.97.3 NAT | 1997 | 2M

For an input signal $4 \sin 10t$, the voltage across the resistance R in the circuit shown in figure, is ... V

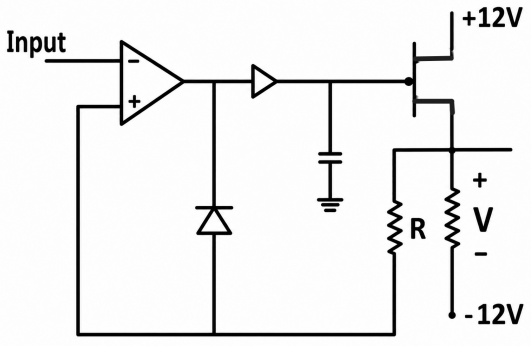


Fig. Q3.97.3

Q3.97.4 NAT | 1997 | 5M

Determine the frequency of oscillation of the circuit shown in figure. Assume the op-amp to be ideal

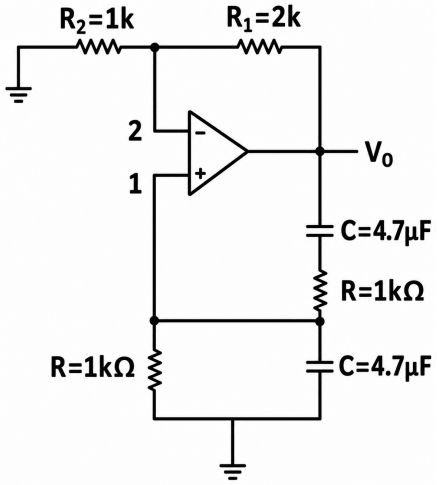


Fig. Q3.97.4

Q3.96.1 MCQ 1996 1M

A non-inverting Op-Amp amplifier is shown in figure. The output voltage V_0 is

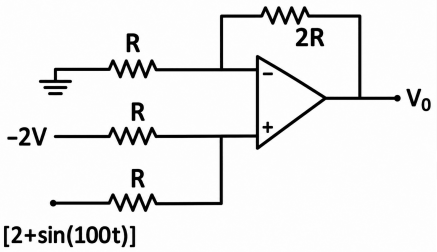


Fig. Q3.96.1

A. $(3/2)\sin(100t)$ **B.** $3\sin(100t)$
C. $2\sin(100t)$ **D.** None of these

Q3.96.2 MCQ 1996 1M

Let the magnitude of the gain in the inverting OP-Amp amplifier circuit shown in be x with switch S_1 open. When the switch S_1 is closed the magnitude of gain becomes

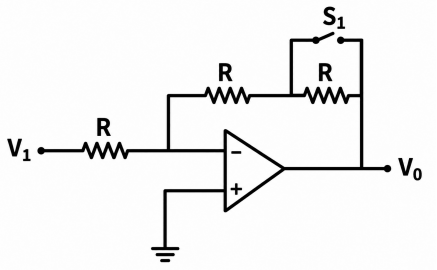


Fig. Q3.96.2

A. x / 2 **B.** -x **C.** 2x **D.** -2x

Q3.95.1 Descriptive 1995 1M

The common mode voltage of a unity gain (voltage follower) op-amp buffer in terms of its output voltage V_0 is

_____

Q3.95.2 NAT 1995 5M

For the circuit shown in figure, determine $\frac{V_0(s)}{V_i(s)}$ and hence write the equations for the magnitude and phase response of $\frac{V_0}{V_i}$. If the value of R_1 is 100 kΩ and of R is 10 kΩ, determine the value of C to obtain a phase shift to 270° between V_0 and V_i for an input frequency of 1000 rad/s.

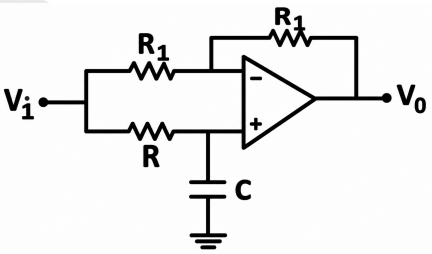


Fig. Q3.95.2

Q3.95.3 Descriptive 1995 5M

For the circuit shown in figure, determine the input impedance Z. assume the op-amp to be an ideal one.

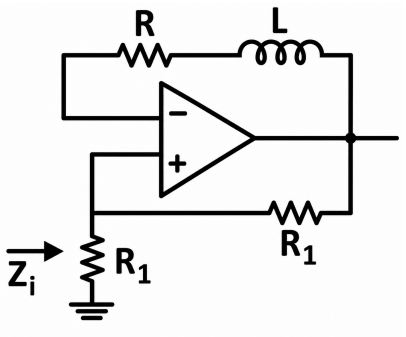


Fig. Q3.95.3

Q3.94.1 Descriptive 1994 1M

An analog comparator is a high-gain amplifier whose output is always either in positive or in negative saturation. (T/F)

_____

Q3.94.2 Descriptive 1994 1M

Given figure shows a two-stage small signal transistor feedback amplifier. Match the defective component (listed on the left hand side below) with its probable effect on the circuit (listed on the right hand side below)

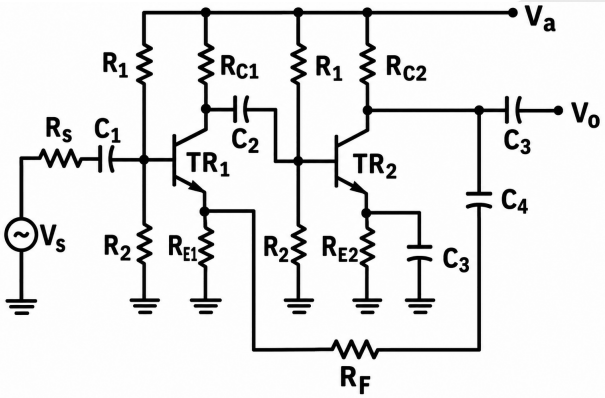


Fig. Q3.94.2

(a) Capacitor C_1 is open
(b) Capacitor C_3 is open
(c) Capacitor C_4 is open
(d) R_{c2} is shorted

(P) All dc voltages normal, V_0 increases marginally
(Q) Collector of TR_2 is at V_{cc}, $V_0 = 0$
(R) All dc voltages normal, gain of 2nd stage decrease V_0
(S) All dc voltages normal, $V_0 = 0$
(T) All dc voltages normal, overall gain of the amplifier increases, V_0 increases
(U) No change

Q3.94.3 NAT 1994 1M

Given figure, shows a non-inverting op-amp summer with $V_1 = 2\,\text{V}$ and $V_2 = -1\,\text{V}$. The output voltage V_0 = _____

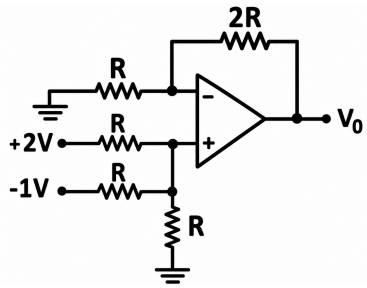


Fig. Q3.94.3

Q3.92.1 MCQ 1992 1M

An ideal OPAMP is used to make an inverting amplifier. The two input terminals of the OP-AMP are at the same potential because

A. The two input terminals are directly shorted internally

B. The input impedance of the OPAMP is infinity

C. The open loop gain of the OPAMP is infinity

D. CMRR is infinity

Q3.92.2 NAT 1992 2M

The circuit shown in figure is excited by the input shown in the figure. Sketch the waveform of the output also indicating the salient values. Assume all components to be ideal.

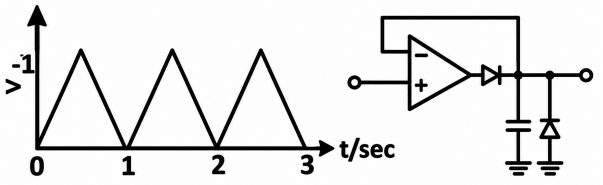


Fig. Q3.92.2

Q3.92.3 NAT 1992 2M

In the following circuit (figure), the output V follows an equation of the form

$$\frac{d^2v}{dt^2} + a\frac{dv}{dt} + bv = f(t)$$

Find a, b and $f(t)$.

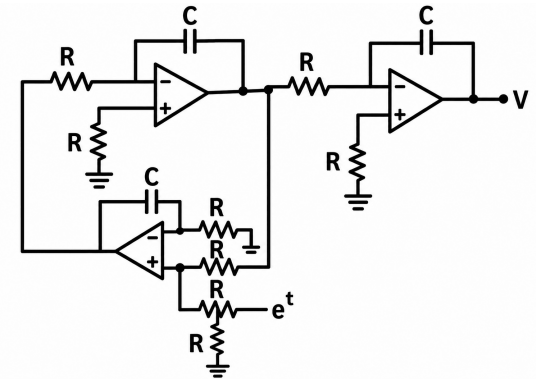


Fig. Q3.92.3

Q3.91.1 NAT 1991

With ideal operational amplifiers, the circuit in figure simulates the equation

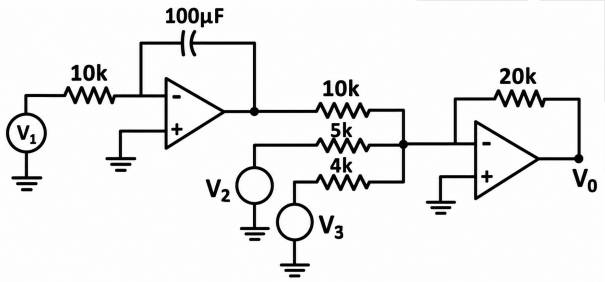


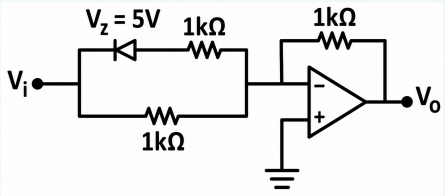
Fig. Q3.91.1

Common Data Questions

The next 2 questions share the data below.

Common data for Q3.91.2 and Q3.91.3

In figure shown, assume the Zener diode and the operational amplifier to be ideal.



Q3.91.2 NAT 1991

Evaluate the gain (V_0 vs V_i) of the circuit for (i) $V_i \leq 0$ (ii) $0 < V_i \leq 5$ V (iii) 5 V $< V_i$

Q3.91.3 NAT 1991

Sketch the gain (V_0 vs V_i) characteristics of the above circuit and label the salient features.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q3.26.1 **A**	Q3.25.1 **D**	Q3.25.2 **2**
Q3.24.1 **39.50 to 41.50**	Q3.23.1 **D**	Q3.22.1 **B**
Q3.22.2 **C**	Q3.22.3 **C**	Q3.21.1 **3150 to 3170**
Q3.20.1 **MTA**	Q3.19.1 **B**	Q3.18.1 **B**
Q3.17.1 **A**	Q3.17.2 **C**	Q3.16.1 **A**
Q3.16.2 **D**	Q3.15.1 **A**	Q3.15.2 **D**
Q3.15.3 **1.1 to 1.4**	Q3.15.4 **A**	Q3.15.5 **A**
Q3.15.6 **B**	Q3.14.1 **D**	Q3.14.2 **B**
Q3.14.3 **0.8**	Q3.14.4 **C**	Q3.13.1 **B**
Q3.13.2 **C**	Q3.12.1 **B**	Q3.11.1 **D**
Q3.11.2 **D**	Q3.10.1 **B**	Q3.09.1 **B**
Q3.09.2 **D**	Q3.09.3 **D**	Q3.08.1 **B**
Q3.08.2 **A**	Q3.08.3 **C**	Q3.08.4 **D**
Q3.07.1 **D**	Q3.07.2 **C**	Q3.06.1 **D**
Q3.06.2 **A**	Q3.05.1 **D**	Q3.05.2 **D**
Q3.04.1 **B**	Q3.04.2 **MTA**	Q3.03.1 **D**
Q3.03.2 **D**	Q3.02.1 **A**	Q3.02.2 $2K\,\Omega, 159.15KHz$
Q3.01.1 **C**	Q3.01.2 **B**	Q3.01.3 **A**

Q3.01.4 0.3 to 0.5	Q3.01.5 *	Q3.00.1 D
Q3.00.2 D	Q3.00.3 *	Q3.99.1 *
Q3.98.1 A-R, B-S, C-P	Q3.98.2 $10, 1, 10^{10}\Omega$	Q3.98.3 *
Q3.97.1 B	Q3.97.2 adder, -3	Q3.97.3 *
Q3.97.4 33.0 to 34.0	Q3.96.1 A	Q3.96.2 A
Q3.95.1 V_0	Q3.95.2 0.1	Q3.95.3 R_1
Q3.94.1 TRUE	Q3.94.2 a-S, b-R, c-T, d-Q	Q3.94.3 1 V
Q3.92.1 C	Q3.92.2 *	Q3.92.3 *
Q3.91.1 *	Q3.91.2 *	Q3.91.3 *

PrepFusion

UNIT

IV

BJTs

Topics

1. BJT Physics and Biasing
2. Current Mirror Circuit Using BJT
3. Small Signal Analysis of BJT Amplifiers
4. Differential Amplifiers and Feedback Topologies Using BJTs

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 29	**MCQ:** 21	**MSQ:** 0	**NAT:** 8
1M: 5	**2M:** 24	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT 2026 1M

In the linear regulator circuit shown, the base to emitter voltage V_{BE} of the BJT is 0.6 V. The Zener diode clamps the base voltage to 5.4 V. Ignore the biasing current of the Zener diode and BJT.

The maximum possible efficiency of the regulator circuit is _____ %. (Round off to one decimal place)

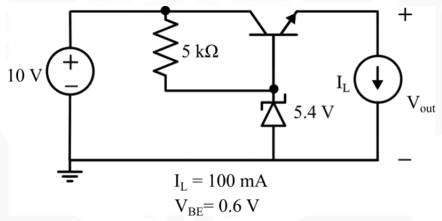


Fig. Q4.26.1

Q4.24.1 MCQ 2024 2M

A BJT biasing circuit is shown in the figure, where $V_{BE} = 0.7$ V and $\beta = 100$. The Quiescent Point values of V_{CE} and I_C are respectively :

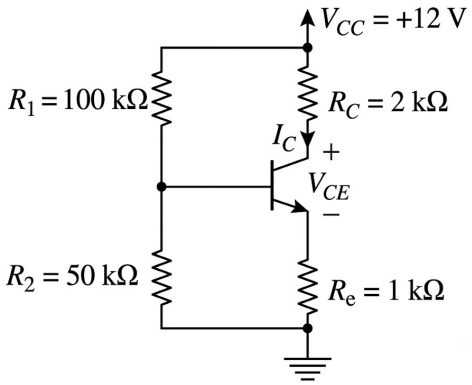


Fig. Q4.24.1

A. 4.6 V and 2.46 mA
B. 3.5 V and 2.46 mA
C. 2.61 V and 3.13 mA
D. 4.6 V and 3.13 mA

Q4.21.1 NAT 2021 1M

In the BJT circuit shown, beta of the PNP transistor is 100. Assume $V_{BE} = -0.7$ V. The voltage across R_C will be 5 V when R_2 is _____ kΩ. (Round off to 2 decimal places)

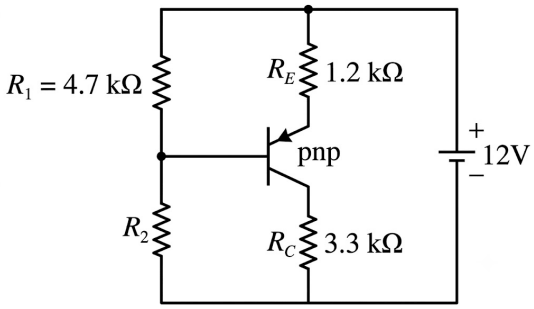


Fig. Q4.21.1

Q4.18.1 NAT 2018 2M

In the circuit shown in the figure, the bipolar junction transistor (BJT) has a current gain $\beta = 100$. The base-emitter voltage drop is a constant, $V_{BE} = 0.7$ V. The value of the Thevenin equivalent resistance R_{TH} (in Ω) as shown in the figure is _____ (up to 2 decimal places).

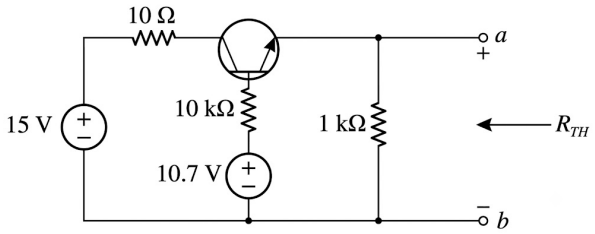


Fig. Q4.18.1

Q4.17.1 NAT 2017 2M

The circuit shown in the figure uses matched transistors with a thermal voltage $V_T = 25$ mV. The base currents of the transistors are negligible. The value of the resistance R in kΩ that is required to provide 1 μA bias current for the differential amplifier block shown is _____. (Give the answer up to one decimal place.)

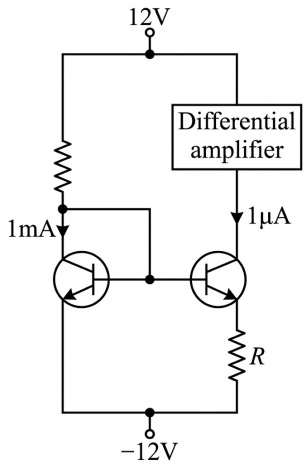


Fig. Q4.17.1

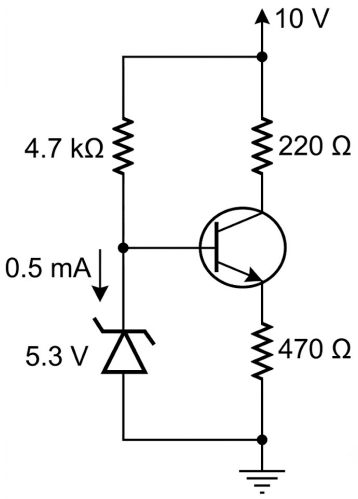


Fig. Q4.16.1

Q4.17.2 MCQ 2017 2M

For the circuit shown in figure below, it is given that $V_{CE} = \frac{V_{CC}}{2}$. The transistor has $\beta = 29$ and $V_{BE} = 0.7\,\text{V}$ when the B-E junction is forward biased.

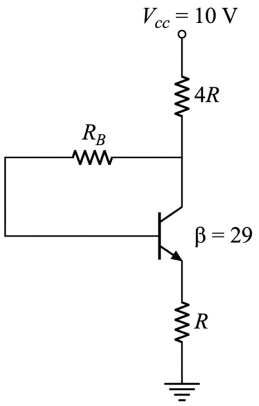


Fig. Q4.17.2

For this circuit, the value of $\frac{R_B}{R}$ is

A. 43 **B.** 92
C. 121 **D.** 129

Q4.16.1 NAT 2016 2M

A transistor circuit is given below. The Zener diode breakdown voltage is 5.3 V as shown. Take base to emitter voltage drop to be 0.6 V. The value of the current gain β is _____.

Q4.15.1 NAT 2015 1M

In the given circuit, the silicon transistor has $\beta = 75$ and a collector voltage $V_C = 9\,\text{V}$. Then the ratio of R_B and R_C is _____.

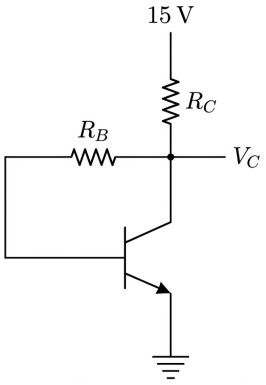


Fig. Q4.15.1

Q4.15.2 NAT 2015 2M

In the following circuit, the transistor is in active mode and $V_C = 2\,\text{V}$. To get $V_C = 4\,\text{V}$, we replace R_C with R'_C. Then the ratio $\frac{R'_C}{R_C}$ is _____.

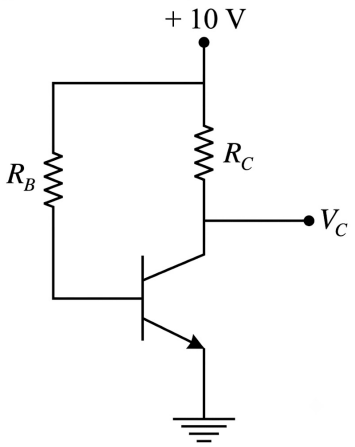


Fig. Q4.15.2

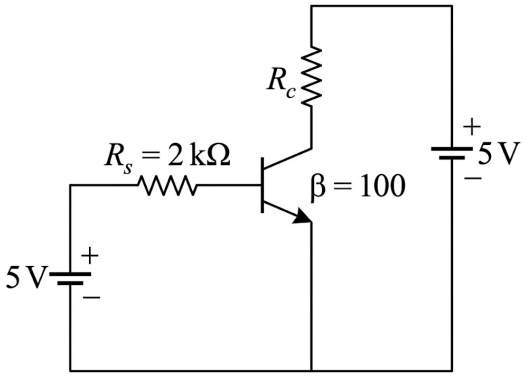


Fig. Q4.14.1

Q4.15.3 MCQ 2015 1M

When a bipolar junction transistor is operating in the saturation mode, which one of the following statements is TRUE about the state of its collector-base (CB) and the base-emitter (BE) junctions?

A. The CB junction is forward biased and the BE junction is reverse biased.

B. The CB junction is reverse biased and the BE junction is forward biased.

C. Both the CB and BE junctions are forward biased

D. Both the CB and BE junctions are reverse biased

Q4.14.1 NAT 2014 2M

The transistor in the given circuit should always be in active region.

Take $V_{CE(sat)} = 0.2\,V$, $V_{BE} = 0.7\,V$

The maximum value of R_c in Ω which can be used, is _____.

Q4.11.1 MCQ 2011 2M

The transistor used in the circuit shown below, has a $\beta = 30$ and I_{CBO} is negligible.

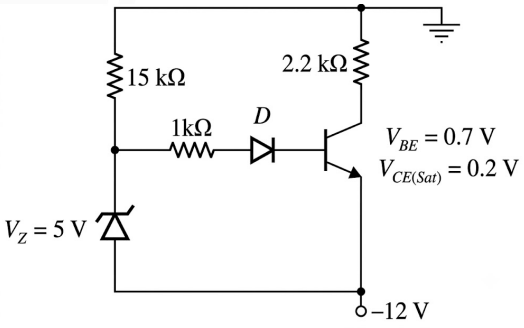


Fig. Q4.11.1

If the forward voltage drop of diode is 0.7 V, then the current through collector will be

A. 168 mA **B.** 108 mA
C. 20.54 mA **D.** 5.36 mA

Q4.10.1 MCQ 2010 2M

The transistor circuit shown uses a silicon transistor with $V_{BE} = 0.7\,V$, $I_C \approx I_E$ and a dc current gain of 100. The value of V_0 is

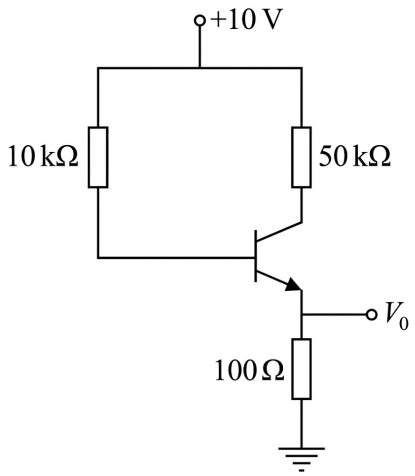


Fig. Q4.10.1

A. 4.65 V
B. 5 V
C. 6.3 V
D. 7.23 V

Q4.08.1 MCQ 2008 2M

Two perfectly matched silicon transistor are connected as shown in the figure. Assuming the β of the transistors to be very high and the forward voltage drop in diodes to be 0.7 V, the value of current I is

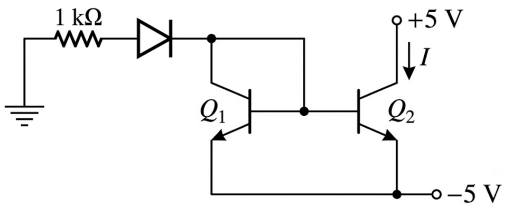


Fig. Q4.08.1

A. 0 mA
B. 3.6 mA
C. 4.3 mA
D. 5.7 mA

Q4.07.1 MCQ 2007 1M

The common emitter forward current gain of the transistor is $\beta_F = 100$.

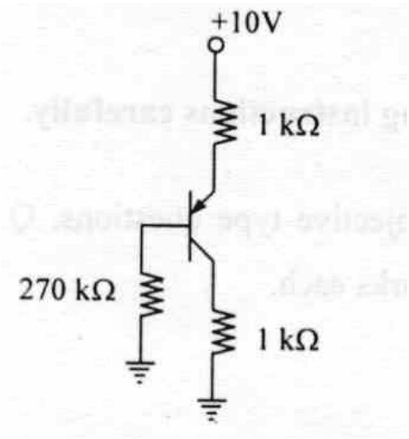


Fig. Q4.07.1

The transistor is operating in

A. Saturation region
B. Cutoff region
C. Reverse active region
D. Forward active region

Q4.07.2 MCQ 2007 2M

The three-terminal linear voltage regulator is connected to a 10 Ω load resistor as shown in the figure. If V_{in} is 10 V, what is the power dissipated in the transistor?

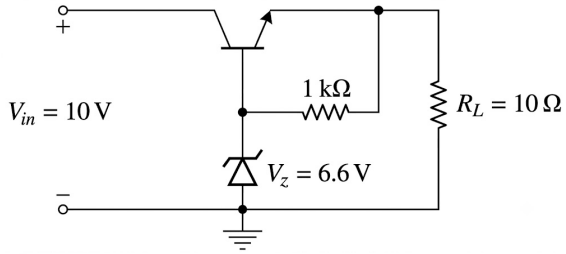


Fig. Q4.07.2

A. 0.6 W
B. 2.4 W
C. 4.2 W
D. 5.4 W

Q4.06.1 MCQ 2006 2M

Consider the circuit shown in figure. If the β of the transistor is 30 and I_{CBO} is 20 nA and the input voltage is +5 V, the transistor would be operating in

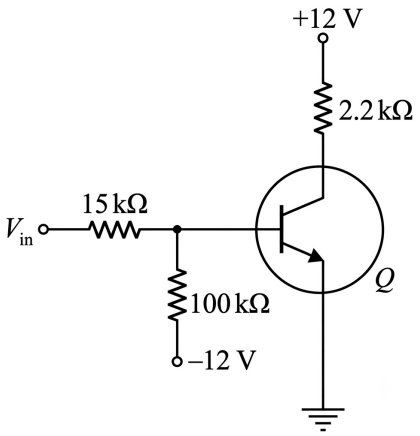


Fig. Q4.06.1

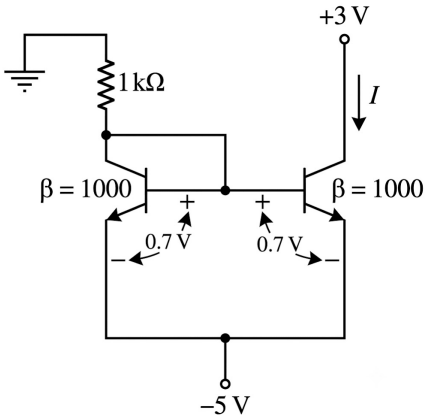


Fig. Q4.04.1

A. Saturation region
B. Active region
C. Breakdown region
D. Cut-off region

Q4.05.1 MCQ 2005 2M

The common emitter amplifier shown in the figure is biased using a 1 mA ideal current source. The approximate value of base current is

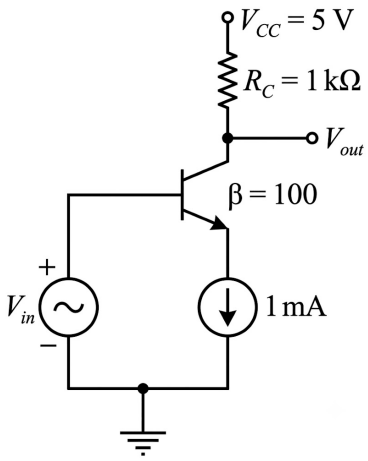


Fig. Q4.05.1

A. 0 μA
B. 10 μA
C. 100 μA
D. 1000 μA

Q4.04.1 MCQ 2004 2M

Two perfectly matched silicon transistors are connected as shown in figure. The value of the current I is

A. 0 mA
B. 2.3 mA
C. 4.3 mA
D. 7.3 mA

Q4.04.2 MCQ 2004 2M

A bipolar junction transistor (BJT) is used as a power control switch by biasing it in the cut-off region (OFF-state) or in the saturation region (ON-state). In the ON state, for the BJT

A. both the base-emitter and base-collector junctions are reverse biased.

B. the base-emitter junctions is reverse biased, and the base-collector junction is forward biased.

C. the base-emitter junction is forward biased, and the base-collector junction is reverse biased.

D. both the base-emitter and base-collector junctions are forward biased.

Q4.04.3 MCQ 2004 2M

In the Schmitt triggered circuit shown in figure, if $V_{CE(sat)} = 0.1$ V, the output logic low level (V_{OL}) is

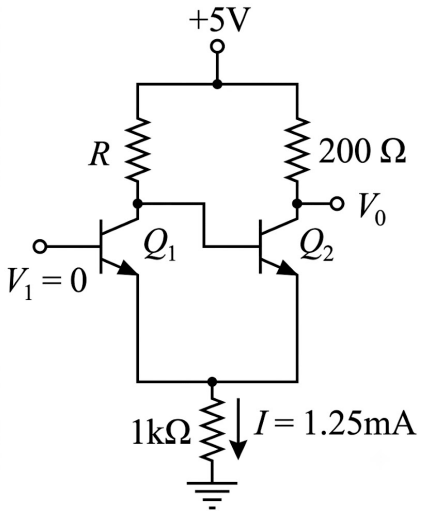


Fig. Q4.04.3

A. 1.25 V
B. 1.35 V
C. 2.50 V
D. 5.00 V

Q4.03.1 MCQ 2003 2M

In the circuit of figure, assume that the transistor has h_{FE} = 99 and V_{BE} = 0.7 V. The value of collector current I_C of the transistor is approximately

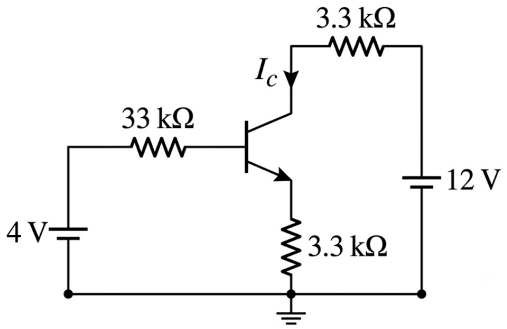


Fig. Q4.03.1

A. [3.3/3.3] mA
B. [3.3/(3.3 + 0.33)] mA
C. [3.3/0.33] mA
D. [3.3/33 + 3.3] mA

Q4.03.2 MCQ 2003 2M

In the circuit shown in figure, the current gain (β) of ideal transistor is 10. The operating point of the transistor (V_{CE}, I_C) is

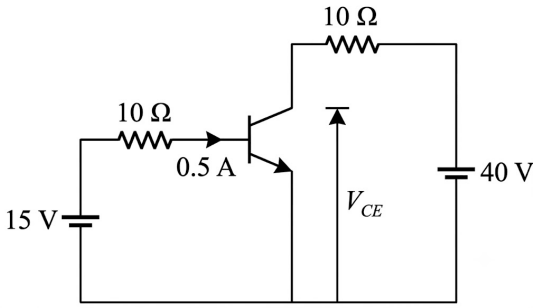


Fig. Q4.03.2

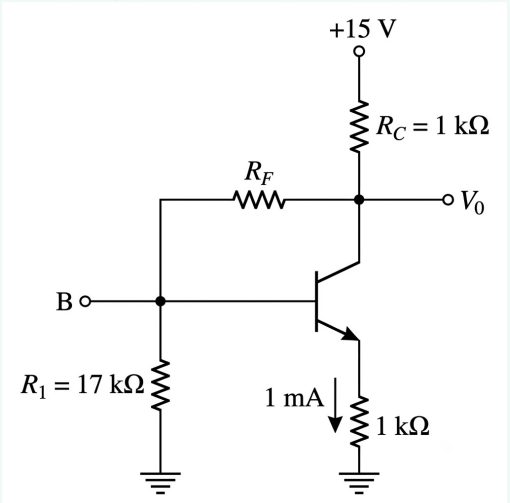
A. 40 V, 4 A
B. 40 V, 5 A
C. 0 V, 4 A
D. 15 V, 4 A

Common Data Questions

The next 3 questions share the data below.

Common Data Note: Answer Questions Q4.02.1, Q4.02.2, and Q4.02.3 using the following data.

For the circuit shown in figure, I_E = 1 mA, β = 99 and V_{BE} = 0.7 V.



Q4.02.1 MCQ 2002 2M

The current through R_C is

A. 0.99 mA
B. 1.1 mA
C. 1.20 mA
D. 1 mA

Q4.02.2 MCQ 2002 2M

Output voltage V_0 will be

A. 16.1 V
B. 14 V
C. 13.9 V
D. None of these

Q4.02.3 MCQ 2002 2M

The value of resistance R_F is

A. 110.9 kΩ
B. 124.5 kΩ
C. 130.90 kΩ
D. 224.5 kΩ

Q4.00.1 MCQ 2000 2M

In the circuit of figure, the value of the base current I_B will be

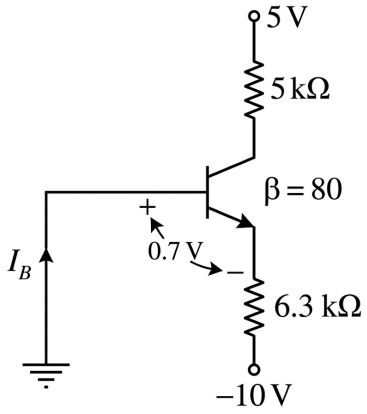


Fig. Q4.00.1

A. 0 μA
B. 18.2 μA
C. 26.7 μA
D. 40 μA

Q4.94.1 MCQ 1994 2M

In the transistor circuit shown in figure, collector to ground voltage is +20 V. Which of the following is the probable cause of error?

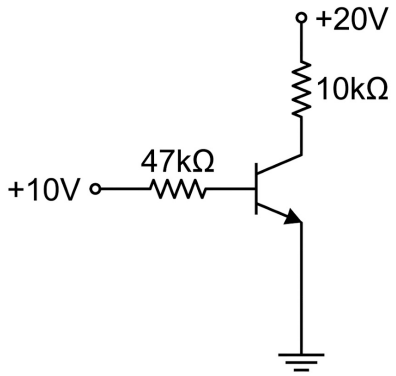


Fig. Q4.94.1

A. Collector emitter terminals shorted.

B. Emitter to ground connection open.

C. 10 kΩ resistor open.

D. Collector base terminals shorted.

Q4.91.1 MCQ 1991 2M

Figure shown below, shows a common emitter amplifier. The quiescent collector voltage of the circuit is approximately

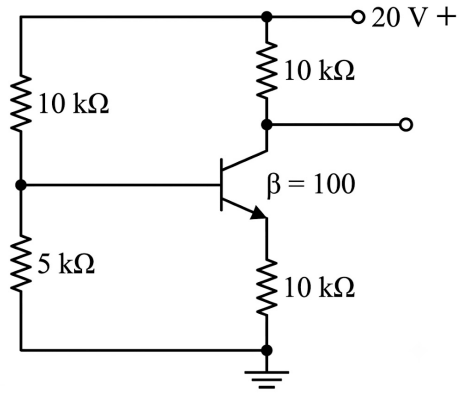


Fig. Q4.91.1

A. 20/3 V
B. 10 V
C. 14 V
D. 20 V

Answer Key

Q4.26.1 47.5 to 48.0	Q4.24.1 A	Q4.21.1 16.70 to 17.70
Q4.18.1 89.00 to 91.00	Q4.17.1 171.0 to 173.0	Q4.17.2 D
Q4.16.1 19	Q4.15.1 104.5 to 106.5	Q4.15.2 0.70 to 0.80
Q4.15.3 C	Q4.14.1 22.00 to 23.00	Q4.11.1 D
Q4.10.1 A	Q4.08.1 B	Q4.07.1 D
Q4.07.2 B	Q4.06.1 B	Q4.05.1 B
Q4.04.1 C	Q4.04.2 D	Q4.04.3 B
Q4.03.1 B	Q4.03.2 C	Q4.02.1 B
Q4.02.2 C	Q4.02.3 A	Q4.00.1 B
Q4.94.1 B	Q4.91.1 C	

UNIT

V

Miscellaneous Topics

Topics

1. CMOS Inverter

Unit Snapshot*

Stream: EE	**Questions:** 1	**MCQ:** 1	**MSQ:** 0	**NAT:** 0
1M: 0	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2026)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 2M

The MOSFET switches shown in the circuit are ideal.

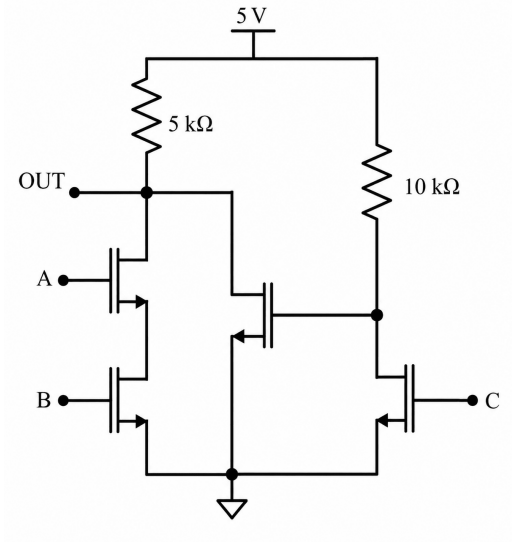


Fig. Q5.26.1

Which of the following is the correct option for Boolean logical expression of the output (OUT), and the maximum possible power (P) consumed by the circuit?

A. OUT = $\overline{\overline{A}\overline{B} + \bar{C}}$, $P = 5$ mW

B. OUT = $\overline{(\overline{A + B})\bar{C}}$, $P = 5$ mW

C. OUT = $\overline{\overline{A}\overline{B}\bar{C}} + \bar{C}$, $P = 7.5$ mW

D. OUT = $\overline{A}\overline{B}C$, $P = 7.5$ mW

Answer Key

Q5.26.1		
D		

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